

claude-windows / c1fcb4aa-537d-41fd-858e-53f93349bf5a

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c1fcb4aa-537d-41fd-858e-53f93349bf5a\subagents\workflows\wf_41c8dc8a-8a4\agent-a1189f99b85c3ad56.jsonl`  
- SHA-256: `1641995456768802b34c5e0a6243131a11192bf5d228d00f00fa67314517f085`  
- Source modified: `2026-07-02T02:53:37+00:00`  
- Imported at: `2026-07-05T16:48:24+00:00`  
- project: `wf_41c8dc8a-8a4`  
- session_id: `c1fcb4aa-537d-41fd-858e-53f93349bf5a`
```

Transcript

1. user (2026-07-02T02:47:11.980Z)

CONTEXT (read carefully):

- You are one auditor in a multi-agent tech-debt/critical-fix audit of the khelsutra-guru multi-repo workspace at C:/Users/avidu/Projects/khelsutra-guru. Today is 2026-07-02. All git repos were just pulled to current origin master/main.
- Repos: badminton-highlight-indexer (the ENGINE, Python, ~1500 tests, org Khelsutra), khelsutra (SPA web/ + marketing site/, TS/React, org avidullu), sports-data-collector, rally-annotator, sports-obsreport (small Python repos). Plus two NON-git dirs: "Annotation Setup", khelsutra-screencast.
- HARD RULES: READ-ONLY. Do NOT modify/create/delete any file, do NOT create branches/worktrees, do NOT pip install, do NOT run the full test suite (CI is trusted on master). You MAY run read-only git and gh commands (gh is authenticated), ruff/mypy IF already installed, and fast read-only scripts.
- EVERY claim must cite file:line (or a gh URL). A claim without file:line evidence is a hypothesis - mark it as such or drop it. The codebase drifts between sessions: re-verify anything you believe from docs against actual code.
- OWNER-GATED areas (flag as owner_gated=true, still report): alpha/serving go-live + Cloudflare dashboard work, Studio P10/P11 (corpus promotion + train/reeval), product/strategy/licensing decisions, real GPU/cloud spend, counsel ToS/DPA, repo rename off "badminton".
- Known-open items from project memory (VERIFY current state, don't trust blindly): engine PR #390 (D1 split main.py, open since 06-26) · audit doc docs/CODE_CLEANUP_AUDIT_2026-06-24.md remaining items D1, D13 (ByteTrack->trackers), Phase-3 (B3/B4/B6/B7/B8/D5/D6/D8-D10/03/04) · docs/DECISION_SNAPSHOT_REGRESSION.md next=P1 · docs/GOLDEN_REGRESSION_FIXTURES.md backlog M4/P0-rename/01/02 · engine issues #340 (Gemini re-bill), #332 (NSFW preflight), #349 (CPU-bound decode), #384 (CI single-source) · khelsutra issues #97-#114 UX/alpha backlog · DO droplet RETIRED 2026-06-25 (docs may still reference it; api.khelsutra.guru CNAME/CF-Access cleanup was pending).
- Categorize each finding: critical-fix (latent bug/correctness/data-loss/cost-leak), security, tech-debt (structure/duplication/dead code), test-gap, ci-infra, doc-rot, hygiene.
- Severity: P0 = latent bug or cost/data/security risk reachable in normal use; P1 = high-value debt blocking velocity or a stale-open thread; P2 = worthwhile cleanup; P3 = nice-to-have (report max 3 of these).
- Be selective: max ~12 findings, each genuinely worth acting on. Note tracked_in when an issue/PR/doc already tracks it (tracked is NOT done - still report). Your final message is data for an orchestrator, not prose for a human.

YOUR DIMENSION: security + configuration debt in the ENGINE. Repo:

C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer.

Check: hardcoded bucket/project/region literals outside the single-source config

(deploy/gcp/buckets.env landed in #410 - grep for khelsutra-rally-corpus, gen-lang-client, hardcoded regions/zones in backend/ deploy/ scripts/ tools/); secret handling (grep for GEMINI_API_KEY, tokens, any literal secrets - also in docs and artifacts); the path-jail + exposure posture (backend/config/startup.py - any gap vs issue #327's scheme-exempt ask); CORS/auth defaults (auth.py, RALLY_CORS_ALLOW_ORIGINS); pip dependency pins + license posture

(pyproject - anything AGPL/unclear per ADR-005); .gitignore correctness (the UTF-16 corruption was fixed in #377 - verify artifacts/ and scratch/ are properly ignored AND untracked).

2. user (2026-07-02T02:47:22.784Z)

```
true
e410136 api: /api/videos surfaces fps + resolution (from the stored ffprobe details) (#413)
81a78bf api: /api/videos returns video_id + a friendly match_name (fix disappearing name) (#412)
8b16cf4 docs: reflect today's storage endpoints + the corpus/user-video boundary (#411)
```

3. user (2026-07-02T02:47:23.703Z)

```
deploy\gcp\ops-agent-startup.sh
deploy\gcp\buckets.env
deploy\gcp\create_gpu_vm.sh
deploy\gcp\README.md
deploy\gcp\provision.sh
deploy\gcp\teardown.sh
```

4. user (2026-07-02T02:47:30.912Z)

```
1      # shellcheck shell=bash
2      # Single source of truth for the GCP deploy layer's PROJECT + BUCKET names.
3      # Every deploy/gcp script `source`s this, so a rename ? or standing up a NEW deployment
/ region ? is
4      # ONE edit here, not a hunt across scripts. Override any value via the environment
before invoking a
5      # script (these are just the defaults; a caller-set env value always wins).
6      #
7      # DATA-FAMILY SEPARATION (never conflate ? this is a compliance boundary, not tidiness):
8      # * TRAINING CORPUS ? golden clips/labels + WASB weights, PERMANENT (no TTL) ->
RALLY_CORPUS_BUCKET
9      # * NIGHTLY/dev OUTPUT ? reports, repo tarball, Cloud Build staging, 30-day TTL ->
RALLY_NIGHTLY_BUCKET
10     # * USER-UPLOADED videos are a SEPARATE surface entirely ? the engine's StorageConfig
11     # (backend/config/models.py) and the per-request `medium_uri` of POST /api/assets ?
and must NEVER
12     # land in these buckets. Keep them physically apart so consent / no-minors / not-
training holds at
13     # the storage layer, not on a promise.
14     #
15     # PER-DEPLOYMENT / PER-REGION: a regional deploy just sets RALLY_GCP_PROJECT +
RALLY_CORPUS_BUCKET
16     # (+ RALLY_NIGHTLY_BUCKET) to its own values (e.g. an EU box -> gs://khelsutra-corpus-
eu). No script
17     # or code change ? that is the whole point of centralizing the names here.
18     #
19     # GCP project every bucket + the worker service account live under.
20     export RALLY_GCP_PROJECT="${RALLY_GCP_PROJECT:-gen-lang-client-0306026826}"
21     #
22     # Training corpus (permanent, no TTL). Canonical name = RALLY_CORPUS_BUCKET. The two
legacy names that
23     # used to be typed independently ? RALLY_GCS_BUCKET (provision.sh) and RALLY_BUCKET
(launch.sh) ? are
24     # still honored for back-compat, then re-exported to the canonical value so any old
invocation OR a
25     # script still reading either name resolves to the SAME bucket (the divergence that
caused drift).
26     export
RALLY_CORPUS_BUCKET="${RALLY_CORPUS_BUCKET:-${RALLY_GCS_BUCKET:-${RALLY_BUCKET:-khelsutra-rally-
corpus}}}"
27     export RALLY_GCS_BUCKET="$RALLY_CORPUS_BUCKET" # back-compat alias (was provision.sh's
variable)
28     export RALLY_BUCKET="$RALLY_CORPUS_BUCKET" # back-compat alias (was launch.sh's
```

```

variable)
29
30     # Nightly / dev EPHEMERAL outputs (30-day TTL; isolated from the corpus so a bucket-wide
TTL can never
31     # touch golden data ? that blast-radius isolation is deliberate).
32     export RALLY_NIGHTLY_BUCKET="${RALLY_NIGHTLY_BUCKET:-khelesutra-rally-nightly}"
33
34     # WASB weights object. Lives UNDER the corpus bucket, so it is derived from
RALLY_CORPUS_BUCKET (a
35     # corpus rename carries the weights path with it); override RALLY_WEIGHTS_URI to point
elsewhere.
36     export RALLY_WEIGHTS_URI="${RALLY_WEIGHTS_URI:-
gs://${RALLY_CORPUS_BUCKET}/models/wasb_badminton_best.pth.tar}"
37

```

5. user (2026-07-02T02:47:34.400Z)

```

deploy\gcp\buckets.env:20:export RALLY_GCP_PROJECT="${RALLY_GCP_PROJECT:-gen-lang-
client-0306026826}"
deploy\gcp\buckets.env:26:export
RALLY_CORPUS_BUCKET="${RALLY_CORPUS_BUCKET:-${RALLY_GCS_BUCKET:-${RALLY_BUCKET:-khelesutra-rally-
corpus}}}"
deploy\gcp\nightly_regression\create_nightly_bucket.sh:5:# gs://khelesutra-rally-corpus ?
PERMANENT, NO TTL: the golden clips + labels + WASB weights
eval_baselines\reelcount_manifest.json:15:      "uri": "gs://khelesutra-rally-corpus/nightly-
inputs/videos/GX030094.mp4",
eval_baselines\reelcount_manifest.json:16:      "labels_uri": "gs://khelesutra-rally-
corpus/nightly-inputs/labels/GX030094_golden_features.json"
eval_baselines\reelcount_manifest.json:20:      "uri": "gs://khelesutra-rally-corpus/nightly-
inputs/videos/Badminton_BXH_2.mp4",
eval_baselines\reelcount_manifest.json:21:      "labels_uri": "gs://khelesutra-rally-
corpus/nightly-inputs/labels/Badminton_BXH_2_golden_features.json"
eval_baselines\reelcount_manifest.json:25:      "uri": "gs://khelesutra-rally-corpus/nightly-
inputs/videos/GX010128.mp4",
eval_baselines\reelcount_manifest.json:26:      "labels_uri": "gs://khelesutra-rally-
corpus/nightly-inputs/labels/GX010128_golden_features.json"
deploy\gcp\README.md:33:| Project | `gen-lang-client-0306026826` (billing **Khelesutra Billing**
`01420D-419EE0-53D83C`) | ? | n/a (shared Gemini project) |
deploy\gcp\README.md:34:| GCS bucket | `gs://khelesutra-rally-corpus` (asia-south1, UBLA) |
~$0.02/GB/mo storage | `gcloud storage rm -r` (? data loss) |
deploy\gcp\README.md:35:| Service account | `rally-worker@gen-lang-
client-0306026826.iam.gserviceaccount.com` (`roles/storage.objectAdmin`) | $0 | `gcloud iam
service-accounts delete` |
deploy\gcp\README.md:51:1. **Service Usage API** ?
https://console.developers.google.com/apis/api/serviceusage.googleapis.com/overview?project=gen-
lang-client-0306026826
deploy\gcp\README.md:52:2. **Link a billing account** (a payment method) ?
https://console.cloud.google.com/billing/linkedaccount?project=gen-lang-client-0306026826
deploy\gcp\README.md:73:gcloud compute project-info describe --project gen-lang-
client-0306026826 \
deploy\gcp\README.md:105: --project=gen-lang-client-0306026826 \
deploy\gcp\README.md:113: --service-account=rally-worker@gen-lang-
client-0306026826.iam.gserviceaccount.com \
deploy\gcp\README.md:132:P=gen-lang-client-0306026826; SA=rally-
worker@$P.iam.gserviceaccount.com
deploy\gcp\README.md:206: "storage": { "backend": "local", "gcs": { "project": "gen-lang-
client-0306026826" } } }
deploy\gcp\README.md:211:# videos -> gs://khelesutra-rally-corpus/videos/ labels ->
gs://khelesutra-rally-corpus/labels/
deploy\gcp\README.md:216: --corpus-uri gcs://khelesutra-rally-corpus --out-uri gcs://khelesutra-
rally-corpus \
deploy\gcp\README.md:221: --video-uri gcs://khelesutra-rally-corpus/videos/GX010135_proxy.mp4 \
deploy\gcp\README.md:222: --out-uri gcs://khelesutra-rally-corpus --segmenter wasb_hybrid
--config config.json
deploy\gcp\README.md:227:- **GCS bucket (recommended, co-located):** `gcs://khelesutra-rally-

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corpus/videos/clip.mp4`. Upload with
deploy\gcp\README.md:228: `gcloud storage cp clip.mp4 gs://khelsutra-rally-corpus/videos/`.
deploy\gcp\README.md:264: `gcloud storage rm -r gs://khelsutra-rally-corpus`.
deploy\gcp\nightly_regression\README.md:30: All management commands run from a machine with
`gcloud` authed to `gen-lang-client-0306026826` (e.g. a
deploy\gcp\nightly_regression\README.md:38: shown). `J=nightly-regression-launcher`, `S=nightly-
regression`, `R=asia-south1`, `P=gen-lang-client-0306026826`.
deploy\gcp\nightly_regression\README.md:63: | `gs://khelsutra-rally-corpus` | **none
(permanent)** | golden **clips + labels** (`nightly-inputs/`) + WASB **weights** (`models/`) |
precious read-only INPUTS ? never auto-deleted |
deploy\gcp\nightly_regression\README.md:87: gcloud storage cp <clip>.mp4 gs://khelsutra-rally-
corpus/nightly-inputs/videos/ # + labels to ../labels/
deploy\gcp\nightly_regression\README.md:98: gcloud run jobs execute nightly-regression-launcher
--region asia-south1 --project gen-lang-client-0306026826
deploy\gcp\nightly_regression\README.md:124: gcloud artifacts docker images delete asia-
south1-docker.pkg.dev/gen-lang-client-0306026826/rally/nightly-launcher --delete-tags
scripts\verify_compute_path.py:15: --video gs://khelsutra-rally-
corpus/videos/mahadevpura_smoke_180s.mp4 \
scripts\verify_compute_path.py:17: --region asia-southeast1 --project gen-lang-
client-0306026826
tests\test_workspace.py:107: cloud = for_video("gs://khelsutra-rally-corpus", VID,
prefix="workspaces")
tests\test_workspace.py:110: rel_cloud = cloud.path(sub)[len("gs://khelsutra-rally-
corpus")]

```

6. user (2026-07-02T02:47:45.071Z)

```

deploy\gcp\teardown.sh:14: # Cross-region VM: set the zone, e.g. RALLY_GCP_ZONE=asia-
southeast1-a bash deploy/gcp/teardown.sh
deploy\gcp\teardown.sh:26: ZONE="${RALLY_GCP_ZONE:-asia-south1-a}"
deploy\gcp\teardown.sh:36: echo "== stopping $VM ? ? the DISK STILL BILLS (~\$/mo); use
'delete' for \$0. Resume: gcloud compute instances start $VM --zone=$ZONE =="
deploy\gcp\teardown.sh:37: gcloud compute instances stop "$VM" --zone="$ZONE" --project
"$PROJECT" || echo " (no running VM named $VM)"
deploy\gcp\teardown.sh:41: gcloud compute instances delete "$VM" --zone="$ZONE" --project
"$PROJECT" --quiet || echo " (no VM named $VM)"
deploy\gcp\teardown.sh:46: gcloud compute instances delete "$VM" --zone="$ZONE" --project
"$PROJECT" --quiet || echo " (no VM)"
deploy\cloudrun\cloudbuild.yaml:11: _REGION: asia-southeast1 # L4 GPU is region-limited
(asia-southeast1 / europe-west1 / europe-west4 / us-central1 / us-east4)
deploy\gcp\provision.sh:2: # Provision the DECOUPLED_COMPUTE GCP storage + IAM (asia-south1 /
Mumbai). Idempotent: safe to
deploy\gcp\provision.sh:15: REGION="${RALLY_GCP_REGION:-asia-south1}"
deploy\gcp\provision.sh:33: echo "== GCS bucket (asia-south1, uniform bucket-level access) =="
deploy\gcp\create_gpu_vm.sh:7: # It sweeps zones for GPU capacity ? asia-south1 (Mumbai, co-
located with the bucket) first, then
deploy\gcp\create_gpu_vm.sh:8: # asia-southeast1 (Singapore), which L4 frequently falls back to
when Mumbai is stocked-out.
deploy\gcp\create_gpu_vm.sh:31: ZONES="${RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-
south1-c asia-southeast1-a asia-southeast1-b asia-southeast1-c}"
deploy\gcp\create_gpu_vm.sh:42: if gcloud compute instances create "$NAME" --project="$PROJECT"
--zone="$Z" \
deploy\gcp\create_gpu_vm.sh:51: echo " SSH: gcloud compute ssh $NAME --zone=$Z"
deploy\gcp\nightly_regression\create_nightly_bucket.sh:21: LOCATION="${RALLY_GCP_LOCATION:-asia-
south1}"
deploy\gcp\nightly_regression\register_scheduler.sh:21: REGION="${RALLY_RUN_REGION:-asia-south1}"
deploy\gcp\nightly_regression\register_scheduler.sh:56: gcloud run jobs deploy "$JOB"
--project="$PROJECT" --region="$REGION" \
deploy\gcp\nightly_regression\register_scheduler.sh:73: gcloud run jobs execute $JOB
--region=$REGION --project=$PROJECT
deploy\gcp\nightly_regression\register_scheduler.sh:74: Watch it: gcloud run jobs executions
list --job=$JOB --region=$REGION --project=$PROJECT
deploy\gcp\nightly_regression\register_scheduler.sh:75: gcloud beta run jobs
executions logs read <execution> --region=$REGION # launcher logs
deploy\gcp\nightly_regression\register_scheduler.sh:76: gcloud compute instances

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get-serial-port-output <vm> --zone=<zone>          # the GPU run
deploy\gcp\nightly_regression\register_scheduler.sh:78: Remove:      gcloud scheduler jobs
delete $SCHED --location=$REGION; gcloud run jobs delete $JOB --region=$REGION
deploy\gcp\nightly_regression\register_scheduler.sh:83: gcloud run jobs execute "$JOB"
--region="$REGION" --project="$PROJECT"
deploy\gcp\nightly_regression\launch.sh:14:# ? The VM-side flow is NOT covered by CI. Tail:
gcloud compute instances get-serial-port-output <name> --zone=<z>.
deploy\gcp\nightly_regression\launch.sh:40:ZONES="{RALLY_GCP_ZONES:-asia-south1-a asia-south1-b
asia-south1-c asia-southeast1-a asia-southeast1-b asia-southeast1-c}"
deploy\gcp\nightly_regression\launch.sh:68: if gcloud compute instances create "$NAME"
--project="$PROJECT" --zone="$Z" \
deploy\gcp\nightly_regression\launch.sh:78: echo "    serial log: gcloud compute instances
get-serial-port-output $NAME --zone=$Z --project=$PROJECT"
deploy\gcp\nightly_regression\launch.sh:79: echo "    if it doesn't self-delete: gcloud
compute instances delete $NAME --zone=$Z -q"
deploy\gcp\nightly_regression\run_on_vm.sh:39: echo "    ? MANUAL CLEANUP REQUIRED: gcloud
compute instances list then ... delete <name> --zone=<zone> -q"
deploy\gcp\nightly_regression\run_on_vm.sh:43: gcloud compute instances delete "$NAME"
--zone="$ZONE" --quiet || \
deploy\gcp\nightly_regression\run_on_vm.sh:44: echo "WARN: self-delete failed ? DELETE
MANUALLY: gcloud compute instances delete $NAME --zone=$ZONE -q"

```

7. user (2026-07-02T02:47:46.453Z)

```

backend\compute\cloud_run.py:127:         token: Optional[str] = None,
backend\compute\cloud_run.py:136:         # The dispatch token keys the done-marker so concurrent
jobs to one bucket don't collide.
backend\compute\cloud_run.py:138:         self._token = token
backend\compute\cloud_run.py:147:         token = self._token or uuid.uuid4().hex
backend\compute\cloud_run.py:148:         done_uri = f"{out_root}/_dispatch/{token}.done"
backend\compute\__init__.py:40:         token: Optional[str] = None,
backend\compute\__init__.py:77:         token=token,
backend\api\auth.py:8:token's `email`, else `sub`).
backend\api\auth.py:33:     """A required Cf-Access token was missing or failed verification ?
the caller returns 401/403."""
backend\api\auth.py:69:def verify_cf_access_token(
backend\api\auth.py:70:    token: str, *, team_domain: str, aud: str, jwks: dict
backend\api\auth.py:86:        kid = jwt.get_unverified_header(token).get("kid")
backend\api\auth.py:88:        raise EdgeAuthError("token header has no kid")
backend\api\auth.py:91:        token,
backend\api\auth.py:105:        raise EdgeAuthError(f"Cf-Access token verification failed:
{exc}") from exc
backend\api\auth.py:109:        raise EdgeAuthError("verified token carries no email/sub claim")
backend\api\auth.py:140:    token = headers.get(CF_ACCESS_HEADER) or
headers.get(CF_ACCESS_HEADER.lower())
backend\api\auth.py:141:    if not token:
backend\api\auth.py:147:    return verify_cf_access_token(token, team_domain=team_domain,
aud=aud, jwks=jwks)
backend\config\models.py:354:    cf_access_aud: str = "" # the CF Access application AUD tag
the token must carry
backend\config\startup.py:24:no token can verify ? **error**, lifting the current first-request
failure to boot), and the
backend\config\startup.py:121:    # (2) Edge auth cannot verify a token without the team domain
+ AUD. Today that only fails on the
backend\config\startup.py:141:    # (3) The auth extra (PyJWT[crypto]) is lazy-imported by
auth.verify_cf_access_token. If it's
backend\assets\fonts\README.md:17:basic Latin, so the Latin brand token ("KheISutra.guru")
renders correctly alongside the local script.
backend\main.py:461:    "gemini_key_present": bool(os.environ.get("GEMINI_API_KEY")),
backend\main.py:1817:    api_key = os.environ.get("GEMINI_API_KEY")
backend\main.py:1818:    if not api_key:
backend\main.py:1821:    "gemini", api_key=api_key, model_name=config.ai_model
backend\reporting\spec.yaml:29:    - { path: "stages.ai_handoff.details.api_key_present", op:
eq, value: false }
backend\reporting\spec.yaml:35:    CREDENTIAL, not 'the video has no rallies'. Fix: put

```

```

GEMINI_API_KEY=<key> in a
backend\reporting\spec.yaml:69:      - { path: "stages.ai_handoff.details.api_key_present", op:
eq, value: true }
backend\storage\gdrive_api.py:16:consumer "connect your Drive" flow: client id/secret + refresh
tokens) is OWNER-gated** ? this backend
backend\storage\gdrive_api.py:205:      page_token: Optional[str] = None
backend\storage\gdrive_api.py:212:      fields="nextPageToken, files(id, name)",
backend\storage\gdrive_api.py:213:      pageToken=page_token,
backend\storage\gdrive_api.py:220:      page_token = resp.get("nextPageToken")
backend\storage\gdrive_api.py:221:      if not page_token:
backend\reporting\harvest.py:158:      key_env = f"{provider.upper()}_API_KEY"
backend\reporting\harvest.py:167:      "api_key_present":
bool(os.environ.get(key_env)) if ai_based else None,
backend\storage\s3.py:9:/ ``AWS_SESSION_TOKEN`` env, instance role, or ``~/.aws``. A ``client``
may be injected
backend\eval\golden_fixtures.py:121:#: / source-path token), so it is rejected up-front by
construction (red-team: free-text id leak).
backend\eval\gemini_refine.py:17:Run (GEMINI_API_KEY in env):
backend\eval\gemini_refine.py:73:      api_key: str,
backend\eval\gemini_refine.py:86:      provider = provider_registry.get("gemini", api_key=api_key,
model_name=model)
backend\eval\gemini_refine.py:138:      api_key = os.environ.get("GEMINI_API_KEY")
backend\eval\gemini_refine.py:139:      if not api_key:
backend\eval\gemini_refine.py:140:          raise SystemExit("GEMINI_API_KEY not set in env")
backend\eval\gemini_refine.py:161:          api_key,
backend\eval\gemini_refine.py:230:      api_key = os.environ.get("GEMINI_API_KEY")
backend\eval\gemini_refine.py:231:      if not api_key:
backend\eval\gemini_refine.py:232:          raise SystemExit("GEMINI_API_KEY not set in env")
backend\eval\gemini_refine.py:256:          "gemini", api_key=api_key, model_name=model
backend\utils\app_home.py:62:      """"``<app-home>/.env`` ? the dotenv file the server prefers
(e.g. ``GEMINI_API_KEY``).""""
backend\providers\__init__.py:16:      def get(self, name: str, api_key: str, model_name: str) ->
AIVideoProvider:
backend\providers\__init__.py:21:          return self._registry[name](api_key=api_key,
model_name=model_name)
backend\ui\assets\index-ukuS9wxa.js:57:[Omitted long matching line]
backend\ui\assets\index-ukuS9wxa.js:60:[Omitted long matching line]
backend\eval\golden_real_fixtures.py:80:#: The fixed schema filenames the writers ALWAYS emit
(as ``paths`` literals). A source-path token
backend\eval\golden_real_fixtures.py:100:      (b) a 32-hex MD5 (the ``video_id`` shape), or (c)
any identifying token of a source path
backend\eval\golden_real_fixtures.py:123:      for token in (sp, p.name, p.stem):
backend\eval\golden_real_fixtures.py:124:          t = (token or "").strip().lower()
backend\eval\golden_real_fixtures.py:125:          # Skip ONLY a token that EXACTLY equals a
schema-literal the writers always emit
backend\eval\golden_real_fixtures.py:127:          # skip on substring containment: a token
like ``label`` or ``traj`` is a SUBSTRING of
backend\eval\golden_real_fixtures.py:134:          f"{where}: source-path token
{token!r} found in written output ? the source "
backend\eval\golden_real_fixtures.py:199:      contain no FORBIDDEN_KEY / MD5 / source-path token
would slip through), so it never reaches the
backend\providers\gemini.py:38:      def __init__(self, api_key: str, model_name: str):
backend\providers\gemini.py:39:          super().__init__(api_key, model_name)
backend\providers\gemini.py:47:          api_key=self.api_key,
backend\utils\redact.py:10:``edge_auth_enabled`` (a boolean), ``cf_access_aud`` (a public id),
or an ML ``max_tokens`` knob.
backend\utils\redact.py:20:#: the ``*_key`` / ``*_token`` suffixes (verified to hit ZERO current
AppConfig fields ? guarded by
backend\utils\redact.py:22:#: ``edge_auth_enabled``, ``cf_access_aud``, ``max_tokens``, or
``tokenizer_name``.
backend\utils\redact.py:26:
r"|access[_-]?token|auth[_-]?token|refresh[_-]?token|session[_-]?token"
backend\utils\redact.py:27:      r"|_key$|_token$|^key$|^token$)",
backend\providers\base.py:6:      def __init__(self, api_key: str, model_name: str):
backend\providers\base.py:7:          self.api_key = api_key

```

```

backend\pipeline\stitcher\watermark_i18n.py:5:their language. The brand token ("KheISutra.guru")
stays Latin; only the surrounding phrase localizes.
backend\pipeline\segmenters\gemini.py:102:         api_key_env_var =
f"{provider_name.upper()}_API_KEY"
backend\pipeline\segmenters\gemini.py:103:         api_key = os.environ.get(api_key_env_var)
backend\pipeline\segmenters\gemini.py:104:         if not api_key:
backend\pipeline\segmenters\gemini.py:105:             from backend.pipeline.segmenters._keyhint
import api_key_hint
backend\pipeline\segmenters\gemini.py:108:         f"{api_key_env_var} is not set ? the
{provider_name} segmenter cannot run. "
backend\pipeline\segmenters\gemini.py:109:         f"Set your API key. " +
api_key_hint(api_key_env_var)
backend\pipeline\segmenters\gemini.py:116:         provider_name, api_key=api_key,
model_name=model_name
backend\pipeline\segmenters\trajectory_hybrid.py:203:         api_key_env_var =
f"{provider_name.upper()}_API_KEY"
backend\pipeline\segmenters\trajectory_hybrid.py:204:         api_key =
os.environ.get(api_key_env_var)
backend\pipeline\segmenters\trajectory_hybrid.py:205:         if not api_key:
backend\pipeline\segmenters\trajectory_hybrid.py:206:             from
backend.pipeline.segmenters._keyhint import api_key_hint
backend\pipeline\segmenters\trajectory_hybrid.py:209:         f"{api_key_env_var}
environment variable is not set! "
backend\pipeline\segmenters\trajectory_hybrid.py:211:         +
api_key_hint(api_key_env_var)
backend\pipeline\segmenters\trajectory_hybrid.py:216:         provider_name,
api_key=api_key, model_name=model_name
backend\pipeline\segmenters\yolo_hybrid.py:81:         api_key_env_var =
f"{provider_name.upper()}_API_KEY"
backend\pipeline\segmenters\yolo_hybrid.py:82:         api_key = os.environ.get(api_key_env_var)
backend\pipeline\segmenters\yolo_hybrid.py:83:         if not api_key:
backend\pipeline\segmenters\yolo_hybrid.py:84:             from
backend.pipeline.segmenters._keyhint import api_key_hint
backend\pipeline\segmenters\yolo_hybrid.py:87:         f"{api_key_env_var} environment
variable is not set! "
backend\pipeline\segmenters\yolo_hybrid.py:89:         + api_key_hint(api_key_env_var)
backend\pipeline\segmenters\yolo_hybrid.py:95:         provider_name, api_key=api_key,
model_name=model_name
backend\pipeline\segmenters\_keyhint.py:10:def api_key_hint(var: str) -> str:
backend\pipeline\segmenters\_keyhint.py:13:         return f'In PowerShell:
$env:{var}="your_api_key_here"'
backend\pipeline\segmenters\_keyhint.py:14:         return f"Set it with: export
{var}=your_api_key_here"

```

8. user (2026-07-02T02:47:56.915Z)

No matches found

9. user (2026-07-02T02:47:57.391Z)

```

1      """Startup checks ? per-profile coherence (M5) + exposure-safety posture (M9 hardening).
2
3      A deployment profile (`truly-local` / `cloud-serving` / `cpu-mock`) implies a
coherent
4      runtime. These checks fail/warn fast at process startup ? a loud signal before any
5      (expensive) job runs ? when the active profile is inconsistent with the environment:
6
7      * cloud-serving with a non-cloud storage backend ? a hard error (the one
config bug we
8      can detect with certainty + zero false-positive risk: a cloud profile that can't
reach a
9      bucket can never serve). This is the README §4.3 "loud error before any job".
10     * a GPU mismatch (e.g. `device='cuda'` on a box where CUDA isn't visible, or
`device='auto'`
11     falling back to CPU) ? a warning, never an error: the probe is best-effort

```

```

(torch may live
12         only in the detector subprocess env ? see ``presets.probe_cuda_available``), and the
detector's
13         own healthcheck is the real authority. A warning surfaces it early without false-
failing.
14         * **creds that can't be confirmed** (no ``GOOGLE_APPLICATION_CREDENTIALS`` / not on
Cloud Run) ?
15         a **warning**: Application Default Credentials can come from the metadata server /
gcloud, which
16         we can't probe offline, so the storage backend's own init is the authority; we just
flag it.
17
18         **Exposure-safety posture (server only, ``include_exposure=True``).** When the engine is
*exposed*
19         behind an edge gate (the owner flips ``security.edge_auth_enabled`` on for the
Cloudflare-Access
20         boundary ? the alpha-shareable "tunnel-to-a-box" topology), the gate is only half the
safety story.
21         A half-configured exposure starts *looking* healthy then leaks / 500s per request, so we
verify the
22         rest of the posture at **boot**: the ``allowed_video_root`` path-jail is set (else an
authenticated
23         tenant could read an arbitrary local file ? **error**), the CF team-domain + AUD are
present (else
24         no token can verify ? **error**, lifting the current first-request failure to boot), and
the
25         ``auth`` extra is importable (else every request 500s ? **warning**). Only the HTTP
**server** passes
26         ``include_exposure=True``; the worker reads ``--video-uri`` (not untrusted tenant
paths), so it is
27         never "exposed" and stays unaffected.
28
29         **DEFAULT-OFF (both dimensions):** with no profile selected (``profile == ""``) the
profile checks are
30         a strict **no-op** ? exactly the pre-M5 behaviour; with ``edge_auth_enabled=False``
(today's default)
31         the exposure checks are a strict **no-op** too. So nothing changes for today's manual-
knob deployments.
32         ""
33
34         from __future__ import annotations
35
36         import logging
37         import os
38         from dataclasses import dataclass
39         from typing import Any, Mapping, Optional
40
41         LEVEL_ERROR = "error"
42         LEVEL_WARN = "warn"
43
44         # Storage backends that count as "a cloud bucket" for cloud-serving coherence.
45         _CLOUD_STORAGE = ("gcs", "s3")
46         # STRONG env signals that Google ADC *might* resolve. Deliberately NOT
GOOGLE_CLOUD_PROJECT /
47         # CLOUDSDK_CONFIG ? those are often set without usable creds (a plain project id / a
config dir
48         # with no active login), so they would suppress the heads-up falsely. (s3 creds are
intentionally
49         # left to boto3's default chain ? no parallel heads-up, by design.)
50         _GCP_ENV_HINTS = ("GOOGLE_APPLICATION_CREDENTIALS", "K_SERVICE", "GCE_METADATA_HOST")
51
52
53         @dataclass(frozen=True)
54         class StartupCheck:
55             """One per-profile finding. ``level`` is ``error`` (fatal) or ``warn`` (heads-

```

```

up)."""
56
57     level: str
58     code: str # stable, greppable code (e.g. "STORAGE_INCOHERENT")
59     message: str
60
61
62     class StartupCheckError(RuntimeError):
63         """Raised by ``enforce_startup_checks`` when an active profile fails a FATAL
check."""
64
65         def __init__(self, failures: list[StartupCheck]):
66             self.failures = failures
67             super().__init__(
68                 "deployment profile startup check failed: "
69                 + "; ".join(f"[{c.code}] {c.message}" for c in failures)
70             )
71
72
73     def _creds_discoverable(env: Mapping[str, str]) -> bool:
74         """True if *some* Google ADC source is hinted by the environment. Best-effort ? a
False here
75         only downgrades to a WARNING, never a hard failure (the storage init is the
authority)."""
76         return any(env.get(k) for k in _GCP_ENV_HINTS)
77
78
79     def _auth_extra_available() -> bool:
80         """True if PyJWT (the ``auth`` extra) is importable ? probed WITHOUT importing it
(find_spec),
81         so this stays cheap + side-effect-free. Module-level so tests can monkeypatch it
deterministically
82         (whether PyJWT is installed in the test env must not change the WARN's presence)."""
83         try:
84             import importlib.util
85
86             return importlib.util.find_spec("jwt") is not None
87         except (
88             Exception
89         ): # pragma: no cover - find_spec only raises on a broken parent package
90             return False
91
92
93     def _exposure_checks(config: Any, env: Mapping[str, str]) -> list[StartupCheck]:
94         """Exposure-safety posture for an *exposed* HTTP server. Gated on
``security.edge_auth_enabled``
95         (default False ? ``[]`` = strict no-op, today's behaviour). When edge auth is ON ?
the "I am
96         exposing this engine behind the CF-Access gate" signal ? the gate alone isn't
enough: verify the
97         rest of the posture at boot so a half-configured exposure fails fast instead of
leaking / 500-ing
98         per request. Config-only (no IO beyond the ``find_spec`` auth-extra probe); ``env``
is the
99         caller's environment view (so the host-allowlist check is testable)."""
100         sec = getattr(config, "security", None)
101         if not sec or not getattr(sec, "edge_auth_enabled", False):
102             return [] # DEFAULT-OFF: edge auth off ? no posture enforcement (byte-identical
to today).
103
104         out: list[StartupCheck] = []
105
106         # (1) The path-jail is load-bearing once authenticated tenants can hit /api/process:
WITHOUT it,
107         # a tenant can point the engine at an ARBITRARY local file (allowed_video_root unset

```

```

= "any local
108     # file", the single-user default). Fatal for an exposed engine.
109     root = getattr(sec, "allowed_video_root", None)
110     if not root or not str(root).strip():
111         out.append(
112             StartupCheck(
113                 LEVEL_ERROR,
114                 "EXPOSED_NO_VIDEO_ROOT",
115                 "security.edge_auth_enabled is ON (engine exposed) but
security.allowed_video_root is "
116                 "unset ? an authenticated tenant could make /api/process read an
arbitrary local file. "
117                 "Set allowed_video_root to a jail directory before exposing the
engine.",
118             )
119         )
120
121     # (2) Edge auth cannot verify a token without the team domain + AUD. Today that only
fails on the
122     # FIRST request (auth.resolve_tenant); lift it to BOOT so a misconfigured gate never
starts
123     # "healthy" then 401s every request.
124     team = (getattr(sec, "cf_team_domain", "") or "").strip()
125     aud = (getattr(sec, "cf_access_aud", "") or "").strip()
126     if not team or not aud:
127         missing = " + ".join(
128             n for n, v in (("cf_team_domain", team), ("cf_access_aud", aud)) if not v
129         )
130         out.append(
131             StartupCheck(
132                 LEVEL_ERROR,
133                 "EXPOSED_EDGE_AUTH_INCOMPLETE",
134                 f"security.edge_auth_enabled is ON but {missing} is not configured ? the
Cf-Access JWT "
135                 "cannot be verified (every request would fail). Set
security.cf_team_domain "
136                 "(e.g. https://<team>.cloudflareaccess.com) and security.cf_access_aud
(the CF Access "
137                 "application AUD tag).",
138             )
139         )
140
141     # (3) The auth extra (PyJWT[crypto]) is lazy-imported by
auth.verify_cf_access_token. If it's
142     # missing, every request 500s at verify time ? a boot WARN beats a per-request 500.
Warning, not
143     # fatal: the auth module's own ImportError is the final authority.
144     if not _auth_extra_available():
145         out.append(
146             StartupCheck(
147                 LEVEL_WARN,
148                 "EXPOSED_AUTH_EXTRA_MISSING",
149                 "security.edge_auth_enabled is ON but PyJWT[crypto] is not importable ?
Cf-Access "
150                 "verification will fail at request time. Install the 'auth' extra (pip
install -e .[auth]).",
151             )
152         )
153
154     # (4) P0c: an exposed engine binds 0.0.0.0, but the Host-header (DNS-rebinding)
guard defaults to
155     # loopback-only when RALLY_ALLOWED_HOSTS is unset (see main._allowed_hosts_from_env)
? it would
156     # then 400 every request the edge forwards with its public Host. Boot WARN beats a
silent 400-all.

```

```

157         if not (env.get("RALLY_ALLOWED_HOSTS", "") or "").strip():
158             out.append(
159                 StartupCheck(
160                     LEVEL_WARN,
161                     "EXPOSED_HOST_ALLOWLIST_LOOPBACK",
162                     "security.edge_auth_enabled is ON but RALLY_ALLOWED_HOSTS is unset ? the
Host-header guard "
163                         "defaults to loopback-only and will 400 forwarded requests. Set
RALLY_ALLOWED_HOSTS to your "
164                         "public host(s) (or '*' when the edge gate validates the host).",
165                     )
166             )
167
168         return out
169
170
171     def run_startup_checks(
172         config: Any,
173         *,
174         cuda_available: Optional[bool] = None,
175         env: Optional[Mapping[str, str]] = None,
176         include_exposure: bool = False,
177     ) -> list[StartupCheck]:
178         """Return the startup findings for ``config``.
179
180         ``cuda_available`` is the caller's best-effort GPU probe (``None`` = not probed ?
GPU checks
181         are skipped; the API server passes ``None``, the worker passes
``presets.probe_cuda_available``).
182         ``include_exposure`` adds the exposure-safety posture (``_exposure_checks``) ? the
HTTP **server**
183         passes ``True``; the worker leaves it ``False`` (it is never "exposed"). Both
dimensions are
184         DEFAULT-OFF: no profile ? no profile checks; ``edge_auth_enabled=False`` ? no
exposure checks.
185         Pure: no IO (beyond the ``find_spec`` auth probe), no torch import ? the caller owns
the GPU probe.
186         """
187         e = os.environ if env is None else env
188         checks: list[StartupCheck] = []
189
190         # Exposure-safety posture ? INDEPENDENT of deployment.profile (a tunnel-to-a-box
exposure often
191         # runs with no profile set), gated on edge_auth_enabled (default-OFF). Server-only.
192         if include_exposure:
193             checks.extend(_exposure_checks(config, e))
194
195         profile = getattr(getattr(config, "deployment", None), "profile", "") or ""
196         if not profile:
197             return checks # no profile ? only the exposure posture (itself off unless edge
auth is on).
198
199         storage_backend = getattr(getattr(config, "storage", None), "backend", "local")
200         indexing = getattr(config, "indexing", None)
201         detector_impl = getattr(indexing, "detector_impl", "auto")
202         device = getattr(getattr(indexing, "wasb", None), "device", "cuda")
203
204         if profile == "truly-local":
205             # truly-local reads bytes from the local FS or a *mounted* Drive ? a cloud
backend here is
206             # an inconsistent (but not fatal) choice worth surfacing.
207             if storage_backend not in ("local", "gdrive"):
208                 checks.append(
209                     StartupCheck(
210                         LEVEL_WARN,

```

```

211         "STORAGE_UNUSUAL",
212         f"profile 'truly-local' usually uses local/gdrive storage, but
storage.backend="
213         f"'{storage_backend}'. Reads will go to a remote backend.",
214     )
215 )
216 # GPU heads-up (warning only ? the detector healthcheck is the authority; the
probe is
217 # best-effort and may be False even when a subprocess detector env has CUDA).
218 if cuda_available is False:
219     if device == "cuda":
220         checks.append(
221             StartupCheck(
222                 LEVEL_WARN,
223                 "CUDA_NOT_VISIBLE",
224                 "device='cuda' but no CUDA GPU is visible to this process. If
the box truly "
225                 "has no GPU the detector healthcheck will hard-fail ? set
indexing.wasb.device="
226                 "'auto' for the CPU fallback (M4).",
227             )
228         )
229     elif device == "auto":
230         checks.append(
231             StartupCheck(
232                 LEVEL_WARN,
233                 "CPU_FALLBACK",
234                 "device='auto' and no CUDA GPU is visible to THIS process ? auto
may resolve "
235                 "to CPU (much slower). If torch lives only in the detector
subprocess env this "
236                 "can be a false alarm; the runner's DeviceContext/healthcheck is
the authority.",
237             )
238         )
239
240 elif profile == "cloud-serving":
241     # The ONE fatal coherence check: a cloud profile that can't reach a bucket can't
serve.
242     if storage_backend not in _CLOUD_STORAGE:
243         checks.append(
244             StartupCheck(
245                 LEVEL_ERROR,
246                 "STORAGE_INCOHERENT",
247                 f"profile 'cloud-serving' requires a cloud storage backend "
248                 f"'{storage_backend}'. "
249                 "A cloud job cannot read/write a local-only backend.",
250             )
251         )
252     elif storage_backend == "gcs" and not _creds_discoverable(e):
253         checks.append(
254             StartupCheck(
255                 LEVEL_WARN,
256                 "CREDS_UNCONFIRMED",
257                 "no GOOGLE_APPLICATION_CREDENTIALS / Cloud Run env detected; relying
on Application "
258                 "Default Credentials. The GCS backend will fail loudly at init if
ADC can't resolve.",
259             )
260         )
261     if device == "cpu":
262         checks.append(
263             StartupCheck(
264                 LEVEL_WARN,

```

```

265         "CLOUD_CPU_DEVICE",
266         "profile 'cloud-serving' runs on a GPU (L4) but
indexing.wasb.device='cpu' ? "
267         "detection will not use the GPU.",
268     )
269 )
270 elif cuda_available is False and device in ("cuda", "auto"):
271     checks.append(
272         StartupCheck(
273             LEVEL_WARN,
274             "CLOUD_CUDA_NOT_VISIBLE",
275             "profile 'cloud-serving' expects a GPU but no CUDA GPU is visible to
this process.",
276         )
277     )
278
279 elif profile == "cpu-mock":
280     # The mock profile is for CI/regression ? a real detector here defeats the
purpose.
281     if detector_impl != "stub":
282         checks.append(
283             StartupCheck(
284                 LEVEL_WARN,
285                 "MOCK_NONSTUB_DETECTOR",
286                 f"profile 'cpu-mock' normally uses detector_impl='stub', but
detector_impl="
287                 f"'{detector_impl}'. A real detector defeats the GPU-free mock.",
288             )
289         )
290
291     return checks
292
293
294 def enforce_startup_checks(
295     config: Any,
296     *,
297     cuda_available: Optional[bool] = None,
298     env: Optional[Mapping[str, str]] = None,
299     raise_on_error: bool = True,
300     include_exposure: bool = False,
301     logger: Optional[logging.Logger] = None,
302 ) -> list[StartupCheck]:
303     """Run the checks, LOG each (errors at ERROR, warnings at WARNING), and ? unless
304     ``raise_on_error=False`` ? raise ``StartupCheckError`` if any FATAL check tripped.
305
306     ``include_exposure`` (server-only) adds the exposure-safety posture. Returns the
full findings
307     list (so a caller can inspect/surface them). A no-op with no logging and no raise
when both
308     dimensions are off (``run_startup_checks`` returns ``[]``)."""
309     log = logger or logging.getLogger(__name__)
310     checks = run_startup_checks(
311         config,
312         cuda_available=cuda_available,
313         env=env,
314         include_exposure=include_exposure,
315     )
316     for c in checks:
317         if c.level == LEVEL_ERROR:
318             log.error("[STARTUP] %s: %s", c.code, c.message)
319         else:
320             log.warning("[STARTUP] %s: %s", c.code, c.message)
321     errors = [c for c in checks if c.level == LEVEL_ERROR]
322     if errors and raise_on_error:
323         raise StartupCheckError(errors)

```

```
324         return checks
325
```

10. user (2026-07-02T02:48:07.196Z)

OPEN | Allow gs:// / s3:// / gdrive-api:// ingest on an exposed (jailed) box ? scheme-exempt recognized storage backends from the path-jail

Today

When ``security.allowed_video_root`` is set (required to expose), ``/api/process`` runs ``validate_input_path`` (local-under-jail) **before** resolving the storage scheme, so ``gs://bucket/x.mp4`` ? ``os.path.realpath`` escapes the jail ? **400**. Bucket ingest is therefore impossible on the exposed alpha box. Pinned by ``tests/test_api_endpoints.py::test_process_hosted_mode_rejects_nonlocal_uri`` (gs/s3/gdrive all 400).

Want

The owner wants to ingest large videos via ``gs://`` on the hosted box (far more convenient than a 500 MB browser upload).

Proposed

Make ``validate_input_path`` / the ``/api/process`` preamble **scheme-aware**: either (a) allow recognized storage schemes (``gs://``, ``s3://``, ``gdrive-api://``) **before** the local-jail check, or (b) apply the jail **only** to local refs (validate after ``storage.resolve_input_ref``, jailing only ``backend == "local"``).

? Must NOT reopen the arbitrary-local-file hole ? the jail must still fully apply to **local** paths (traversal/symlink/absolute). Add tests for both the allow (bucket URI accepted) and the reject (local traversal still 400) paths. Pairs with the ``gs://`` ingest the alpha CUJ + DEPLOY_ALPHA \$7 currently disclaim.

11. user (2026-07-02T02:48:08.152Z)

```
backend\main.py:39:from fastapi.middleware.cors import CORSMiddleware
backend\main.py:43:from starlette.middleware.trustedhost import TrustedHostMiddleware
backend\main.py:110:    """M9: CORS allow-origins from the ``RALLY_CORS_ALLOW_ORIGINS`` env var
(comma-separated),
backend\main.py:114:    raw = (src.get("RALLY_CORS_ALLOW_ORIGINS", "") or "").strip()
backend\main.py:119:# CORS for the web UI. allow_origins='' with allow_credentials=True is
rejected by browsers per the
backend\main.py:124:    CORSMiddleware,
backend\main.py:125:    allow_origins=_cors_origins_from_env(),
backend\main.py:132:def _allowed_hosts_from_env(env: Optional[Dict[str, str]] = None) ->
List[str]:
backend\main.py:143:    # IPv4 loopback (the default bind) + the TestClient host. NB:
Starlette's TrustedHostMiddleware
backend\main.py:153:    TrustedHostMiddleware,
backend\main.py:154:    allowed_hosts=_allowed_hosts_from_env(),
backend\config\startup.py:155:    # loopback-only when RALLY_ALLOWED_HOSTS is unset (see
main._allowed_hosts_from_env) ? it would
backend\config\models.py:355:    # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS
env var ? read at import in
```

12. user (2026-07-02T02:48:13.796Z)

```
100     ``request.app.state`` exclusively."""
101     global global_config, db_instance
102     global_config = config
103     db_instance = db
104     app.state.config = config
105     app.state.db = db
106     return app
107
108
```

```

109     def _cors_origins_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
110         """M9: CORS allow-origins from the ``RALLY_CORS_ALLOW_ORIGINS`` env var (comma-
separated),
111         default ``["*"]``. Read from the ENV (not a cwd ``config.json``) so importing
``backend.main``
112         does zero filesystem I/O ? a hosted/multi-tenant deploy sets this to lock CORS to
its origin."""
113         src = os.environ if env is None else env
114         raw = (src.get("RALLY_CORS_ALLOW_ORIGINS", "") or "").strip()
115         origins = [o.strip() for o in raw.split(",") if o.strip()]
116         return origins or ["*"]
117
118
119     # CORS for the web UI. allow_origins='' with allow_credentials=True is rejected by
browsers per the
120     # fetch spec + is unsafe to carry into multi-tenant; this tool uses no cookies, so
credentials stay
121     # off. The origin list is configurable via the env var above (default ["*"]) so a hosted
deploy can
122     # lock it to its origin; the middleware is fixed at startup.
123     app.add_middleware(
124         CORSMiddleware,
125         allow_origins=_cors_origins_from_env(),
126         allow_credentials=False,
127         allow_methods=["*"],
128         allow_headers=["*"],
129     )
130
131
132     def _allowed_hosts_from_env(env: Optional[Dict[str, str]] = None) -> List[str]:
133         """P0c: Host-header allowlist ? a DNS-rebinding guard for the localhost appliance. A
malicious
134         web page can point a domain it controls at 127.0.0.1 and script requests to the
local server;
135         the browser sends ``Host: attacker.com``, which this rejects (the server only
answers loopback
136         Host headers by default). ``RALLY_ALLOWED_HOSTS`` (comma-separated) overrides; the
default keeps
137         Starlette's TestClient host ``testserver`` so the in-process suite isn't broken.
Read from the
138         ENV (not config.json) so importing ``backend.main`` does zero filesystem I/O; a
routable/edge
139         deploy sets its public host (or ``*`` when fronted by an auth gate that validates
the host)."""
140         src = os.environ if env is None else env
141         raw = (src.get("RALLY_ALLOWED_HOSTS", "") or "").strip()
142         hosts = [h.strip() for h in raw.split(",") if h.strip()]
143         # IPv4 loopback (the default bind) + the TestClient host. NB: Starlette's
TrustedHostMiddleware
144         # matches on host.split(":")[0], which mangles a bracketed IPv6 Host ("[::1]:port"),
so IPv6
145         # loopback is NOT supported by this guard ? access the appliance via
127.0.0.1/localhost.
146         return hosts or ["localhost", "127.0.0.1", "testserver"]
147
148
149     # DNS-rebinding guard. Bound to localhost by default, so only loopback Host headers are
honoured
150     # (a remote page can't read responses cross-origin AND its Host header is rejected
here). www_redirect
151     # off so an unexpected ``www.`` host is rejected rather than 307-redirection.
152     app.add_middleware(
153         TrustedHostMiddleware,
154         allowed_hosts=_allowed_hosts_from_env(),
155         www_redirect=False,

```

```

156     )
157
158
159     def _resolve_bind_host(
160         config: Optional[AppConfig] = None, cli_host: Optional[str] = None
161     ) -> str:
162         """P0c: the server bind address. Precedence: ``--host`` > ``RALLY_BIND_HOST`` env >
163         ``0.0.0.0``
164         when an edge-auth profile is active (``security.edge_auth_enabled`` ? i.e. there IS
165         an auth gate
166         in front) > ``127.0.0.1`` (default: localhost-only, today's behaviour). Binding a
167         routable
168         interface with no auth gate is unsafe, hence the localhost default; widen it
169         consciously."""
170         if cli_host and cli_host.strip():
171             return cli_host.strip()
172         env = os.environ.get("RALLY_BIND_HOST")
173         if env and env.strip():

```

13. user (2026-07-02T02:48:14.151Z)

```

1     # PEP 621 packaging for the badminton/sports highlight indexer (DEFERRED #7).
2     #
3     # Build backend: hatchling. `backend/` is an intentional NAMESPACE package
4     # (no `backend/__init__.py`; mypy.ini sets explicit_package_bases for the same
5     # reason), so the wheel target packages are listed EXPLICITLY below ? hatchling's
6     # auto-detection would otherwise miss the un-__init__'d `backend` root.
7     #
8     # torch/torchvision are deliberately NOT listed in [project.dependencies]:
9     # their CPU/CUDA wheels need an out-of-band index (--index-url / --extra-index-url),
10    # which PEP 621 cannot express. Install them separately ? see [project.optional-
11    dependencies]
12    # `gpu` (documentation only) and the README. CI installs CPU torch on its own line.
13    [build-system]
14    requires = ["hatchling"]
15    build-backend = "hatchling.build"
16
17    [project]
18    name = "badminton-highlight-indexer"
19    version = "0.1.0"
20    description = "AI-assisted rally segmentation and highlight indexing for racket sports."
21    readme = "README.md"
22    requires-python = ">=3.10"
23    license = { text = "MIT" }
24    authors = [{ name = "Avi Dullu", email = "avi.dullu@gmail.com" }]
25    keywords = ["badminton", "sports", "video", "highlights", "rally", "segmentation"]
26
27    # Runtime deps mirror requirements.txt, EXCEPT torch/torchvision (see header).
28    dependencies = [
29        "fastapi>=0.111.0",
30        "uvicorn>=0.30.1",
31        "opencv-python>=4.9.0.80",
32        "numpy>=1.26.4",
33        "pydantic>=2.7.1",
34        "jinja2>=3.1.4",
35        "python-dotenv>=1.0.0",
36        "google-genai>=2.4.0",
37        "supervision>=0.21.0",
38        # Used directly on the serving path (backend/features/rally_seq.py: uniform_filter1d
39        /
40        # maximum_filter1d); previously present only TRANSITIVELY via supervision, so a
41        supervision
42        # bump or a pruned-transitive freeze could silently break inference. Declare it.
43        (BSD-3.)
44        "scipy>=1.10",

```

```

41     # D13/P12: ByteTrack moved to the standalone `trackers` package (Apache-2.0). Its
42     # PyPI wheel is broken (metadata-only), so it MUST be installed from git ? see
43     # requirements.txt. PEP 621 cannot express a git URL, so it's NOT listed here
44     # (same pattern as torch/torchvision). CI installs it on its own line.
45 ]
46
47 [project.optional-dependencies]
48 # Test + lint/type tooling ? matches the CI lanes (.github/workflows/ci.yml).
49 dev = [
50     "pytest",
51     "pytest-cov",
52     "httpx",          # fastapi TestClient transport
53     "ruff",
54     "mypy",
55     "PyJWT[crypto]>=2.8", # M9 edge-auth: Cf-Access JWT verify (RS256) ? tests + CI
56 ]
57
58 # M9 edge-auth (Cloudflare Access). DEFAULT-OFF: only needed when
security.edge_auth_enabled=True.
59 # `jwt` is lazy-imported, so the base runtime works without this; install it for
hosted/multi-tenant.
60 auth = ["PyJWT[crypto]>=2.8"]
61
62 # Per-video observability reports (SOFT dependency: the pipeline degrades to a
63 # no-op if absent). Pinned to its git tag per BACKLOG. The private repo is
64 # fetched over SSH; if you lack access, simply omit this group ? reporting
65 # tests skip via pytest.importorskip. For local dev against a sibling checkout:
66 #   pip install -e ../sports-obsreport --no-build-isolation
67 reporting = [
68     "obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3",
69 ]
70
71 # `gpu` is DOCUMENTATION ONLY ? it is intentionally empty. torch/torchvision
72 # CUDA wheels require `--index-url`/`--extra-index-url`, which PEP 621 cannot
73 # express, so they are NOT declarable here. Install them out-of-band instead:
74 #   # CUDA 12.4 box:
75 #   pip install torch>=2.12.0 torchvision>=0.27.0 --index-url
https://download.pytorch.org/whl/cu124
76 #   # CPU-only box (also what CI uses):
77 #   pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
78 gpu = []
79
80 # Storage backends (docs/DECOUPLED_COMPUTE/). The default `local` backend needs NONE of
81 # these. Cloud SDKs are imported lazily, so `import backend.storage` works without them;
82 # install the extra for the bucket you use. (boto3's S3 client covers AWS + Cloudflare
R2 +
83 # DO Spaces + MinIO via endpoint_url.) `gdrive://` needs NO extra ? it reads a locally
84 # MOUNTED Google Drive (Google Drive for Desktop) as plain files; see
backend/storage/gdrive.py.
85 storage-s3 = ["boto3>=1.34"]
86 storage-gcs = ["google-cloud-storage>=2.16"]
87
88 [project.scripts]
89 # `indexer --server`, `indexer --help`, etc. -> backend/main.py::main()
90 indexer = "backend.main:main"
91
92 [tool.hatch.metadata]
93 # The `reporting` extra pins obsreport to a git tag (a PEP 508 *direct
94 # reference*); hatchling requires opting in to allow those in metadata.
95 allow-direct-references = true
96
97 [tool.hatch.build.targets.wheel]
98 # `backend` is a namespace package (no __init__.py); list both import roots
99 # explicitly so an editable/wheel install exposes `import backend` and
100 # `import training` regardless of cwd.

```

```
101     packages = ["backend", "training"]
102
```

14. user (2026-07-02T02:48:30.568Z)

```
README.md:20:[Omitted long matching line]
.github\workflows\nightly.yml:165: # copyleft (GPL/AGPL/LGPL/SSPL). This is exactly the
transitive-dep drift the
.github\workflows\nightly.yml:183: - name: Fail on any copyleft (GPL/AGPL/LGPL/SSPL) in the
runtime closure
.github\workflows\nightly.yml:186: echo "::error::A runtime dependency carries a
copyleft (GPL/AGPL/LGPL/SSPL) license ? the guardrail is MIT/Apache/BSD only. Review the report
above."
.github\workflows\nightly.yml:189: echo "OK: no GPL/AGPL/LGPL/SSPL in the runtime
dependency closure."
.github\workflows\ci.yml:178: # (GPL/AGPL/LGPL/SSPL). torch (BSD-3) is installed out-of-band
and is not in this closure. This
.github\workflows\ci.yml:199: - name: Fail on any copyleft (GPL/AGPL/LGPL/SSPL) in the
runtime closure
.github\workflows\ci.yml:202: echo "::error::A runtime dependency carries a copyleft
(GPL/AGPL/LGPL/SSPL) license ? the guardrail is MIT/Apache/BSD only. Review the report above."
.github\workflows\ci.yml:205: echo "OK: no GPL/AGPL/LGPL/SSPL in the runtime dependency
closure."
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md:69: numpy STFT, so the originally-
anticipated scipy dependency was not added. Avoid heavy/AGPL audio stacks. Any
docs\archives\decisions\DECISIONS.md:239:[Omitted long matching line]
docs\archives\decisions\DECISIONS.md:243:[Omitted long matching line]
docs\archives\decisions\DECISIONS.md:251:- Detector swap is contained to
`backend/indexer/yolo_hybrid.py` (the `from ultralytics import YOLO` load + the
`model.track(...)` call) plus `config.json` knobs and `tests/test_yolo_hybrid.py` mocks. Stage-1
feature outputs (velocity/active-frame/track-id stats) and Stages 2?4 are detector-agnostic and
unchanged.
docs\archives\decisions\DECISIONS.md:252:- We now own the frame loop + tracking (ultralytics did
both internally). Use a permissive ByteTrack (e.g. `supervision`, MIT, or the MIT ByteTrack
repo) on top of `lapx` (already a BSD dep).
docs\archives\decisions\DECISIONS.md:253:- Residual items needing legal sign-off: exact codec-
pool royalty thresholds (AVC/HEVC decode of uploads), Ultralytics Enterprise pricing, and per-
checkpoint weight licenses ? tracked in `docs/MONETIZATION_AUDIT.md` §7.
docs\archives\MONETIZATION_AUDIT.md:7:> forward. (Notably: the AGPL/Ultralytics blocker in §1 is
now resolved via ADR-005.)
docs\archives\MONETIZATION_AUDIT.md:12:**Key legal premise (verified):** We do **not**
distribute code or binaries to users; everything runs server-side. Therefore *distribution-
triggered* copyleft (GPL/LGPL) is **not** triggered. BUT **AGPL §13** (network/SaaS clause) and
**external metered-API terms** still apply, and **codec patent royalties** are a separate axis
from software licenses.
docs\archives\MONETIZATION_AUDIT.md:22:| **ultralytics** (lib) | **AGPL-3.0** | **Yes** ? §13
network clause; importing the lib into our app + serving over a network = source-disclosure of
the *whole* app | ? **BLOCKER** | Replace (see §2) or buy Enterprise License |
docs\archives\MONETIZATION_AUDIT.md:23:| **`yolov8n.pt`** weights | **AGPL-3.0** (same as code;
HF card tagged `agpl-3.0`) | **Yes** | ? **BLOCKER** | Replace weights; swapping codebase but
keeping `.pt` does *not* escape |
docs\archives\MONETIZATION_AUDIT.md:28:| **ffmpeg / ffprobe** | LGPL-2.1+ (GPL if `--enable-
gpl`); **not AGPL** | **No** ? no distribution, no network clause | ? CLEAR | Keep; invoke as
subprocess (already do) |
docs\archives\MONETIZATION_AUDIT.md:36:**Bottom line:** Only **two** real blockers ? (1)
**Ultralytics** (AGPL, code *and* weights) and (2) two specific **model weights** (MonoTrack,
YOLO-NAS) we must simply avoid. Everything else is either already clean or a manageable WATCH.
docs\archives\MONETIZATION_AUDIT.md:44:### Player / motion-gating detector (replaces
`ultralytics` YOLOv8n)
docs\archives\MONETIZATION_AUDIT.md:51:| **RT-DETRv2** (`lyuwenyu`/Baidu) | **Apache-2.0** |
Real-time, NMS-free transformer; competitive with YOLO. Use the **Baidu** repo, *not* the
ultralytics RT-DETR port (that inherits AGPL). |
docs\archives\MONETIZATION_AUDIT.md:52:| **D-FINE** | **Apache-2.0** core + weights | ICLR'25
SOTA real-time. ?? repo bundles an **AGPL YOLO** subcomponent ? use only the Apache RT-DETR-
derived core. |
```

docs\archives\MONETIZATION_AUDIT.md:71:- **Ultralytics Enterprise License** ? removes AGPL for closed-source/SaaS use of code **and** models. Pricing is **quote-only / not public** (confirmed via Ultralytics GH issues #15877, #1888, #9853 ? all routed to sales) ?? Treat as a material recurring cost; only sensible if YOLOv8 accuracy proves irreplaceable.

docs\archives\MONETIZATION_AUDIT.md:110:1. **Ultralytics Enterprise price** ? get a quote to compare against swap engineering cost.

docs\archives\MONETIZATION_AUDIT.md:118:- Ultralytics License (AGPL-3.0 + Enterprise):
`ultralytics.com/license`, `docs.ultralytics.com/help/license/`, repo `LICENSE` (GitHub API `spdx\_id=AGPL-3.0`), weights tag on HF `Ultralytics/YOLOv8` (`agpl-3.0`), GH issues #15877/#1888/#9853.

docs\archives\MONETIZATION_AUDIT.md:119:- AGPL-3.0 §13 text: `gnu.org/licenses/agpl-3.0`.

docs\archives\MONETIZATION_AUDIT.md:123:- Detectors: `Megvii-BaseDetection/YOLOX` (Apache), `lyuwenyu/RT-DETR` (Apache), `Peterande/D-FINE` (Apache + AGPL YOLO caveat), `Deci-AI/super-gradients` `LICENSE.YOLONAS.md` (noncommercial weights).

docs\ROADMAP.md:25:> AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +

docs\ROADMAP.md:27:> player-motion gating stage, not the Ultralytics library.

docs\MULTI_SIGNAL_FUSION_PLAN.md:231:Every dependency stays **MIT / Apache-2.0 / BSD** (ADR-002 / ADR-005; `ultralytics` was dropped for

docs\MULTI_SIGNAL_FUSION_PLAN.md:232:AGPL-3.0).

docs\MULTI_SIGNAL_FUSION_PLAN.md:241:| **Banned** | **Ultralytics YOLOv5/8/11 (AGPL-3.0)** .
MonoTrack (Adobe, non-comm) . **YOLO-NAS** (Deci, non-comm) | ? | ? |

docs\MULTI_SIGNAL_FUSION_PLAN.md:297: (no `ultralytics` import; detector factories stay torchvision/Apache).

docs\PACKAGING_PLAN.md:55:- **New packaging deps that clear L6** `[researched]`:**uv** (Apache/MIT), Inno Setup (permissive), `onnxruntime` (MIT), `micromamba` (BSD-3), Docker/OCI (Apache-2.0), boto3/google-cloud-* (Apache-2.0), stdlib `webbrowser`. **Forbidden:** `pystray` (LGPL-3.0). **Build-time-only:** `Nuitka` 4.x is **AGPL-3.0+** (compiled output has a runtime exception) ? usable as a tool, never bundled. `obsreport` (private git dep) stays a truly-optional extra, never bundled.

docs\PACKAGING_PLAN.md:103:| P0d | **Declare `scipy`** in `[project.dependencies]` (serving use `[verified features/rally\_seq.py:17]`, was transitive-only) + **license-scan CI lane** (pip-licenses; fails on GPL/AGPL/LGPL/SSPL) + **built-wheel smoke lane** (build ? install ? `indexer --help` from a non-repo cwd). Both lanes green on CI. | ? | No | ? |

[#297](https://github.com/Khelsutra/badminton-highlight-indexer/pull/297) |

docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md:38: GRPO/RFT with a custom video reward** in one stack. Keep **unsloth** (Apache-2.0 core; *avoid its AGPL Studio UI*)

docs\CODE_MAP.md:161:[Omitted long matching line]

docs\BACKLOG.md:31:- ? **Ultralytics Enterprise pricing** ? get a quote **only if** the torchvision/ByteTrack detector (PR #8) proves insufficient on real footage. Currently **not needed** (swap done). Quote-only pricing. **Revisit:** if detector quality forces reconsidering YOLOv8.

docs\COMMERCIALIZATION.md:22:**AGPL §13** (network clause), **metered-API terms**, and **video-codec patent**

docs\COMMERCIALIZATION.md:50:| C1 | **AGPL / copyleft removal** | ? | ? | ADR-005 shipped: AGPL `ultralytics` YOLOv8 replaced with **torchvision Faster R-CNN** (BSD) + `supervision` ByteTrack (MIT). No AGPL code/weights in the stack. | Done. Keep new deps permissive. |

docs\COMMERCIALIZATION.md:62:| ~`ultralytics` + `yolov8n.pt`~ | AGPL-3.0 | Yes (§13) | ?
REMOVED | Replaced per ADR-005 ? no longer in the stack. |

docs\COMMERCIALIZATION.md:72:**Bottom line:** the two hard blockers from the audit (Ultralytics; banned weights)

docs\COMMERCIALIZATION.md:190:1. **Ultralytics Enterprise price** ? only relevant if a permissive detector ever proves

docs\COMMERCIALIZATION.md:208: audit + the commercial-readiness review. **Refreshed** since the audit:**C1 AGPL**

backend\pipeline\detectors\person_detector.py:28:(ADR-005; ultralytics' AGPL YOLO was dropped).

backend\pipeline\detectors\person_detector.py:41:# class 1 (index 0 is the implicit background).

NB: ultralytics YOLO used class 0 ?

backend\pipeline\detectors\person_detector.py:140: This replaces the old ultralytics
`model.track(...)` call, which bundled

backend\pipeline\segmenters\yolo_hybrid.py:9:Stage 1 originally used the AGPL-licensed ultralytics YOLO. To keep the hosted

15. user (2026-07-02T02:48:31.105Z)

1 fastapi>=0.111.0

```

2      uvicorn>=0.30.1
3      opencv-python>=4.9.0.80
4      numpy>=1.26.4
5      pydantic>=2.7.1
6      jinja2>=3.1.4
7      python-dotenv>=1.0.0
8      google-genai>=2.4.0
9      # torch/torchvision are listed here but deliberately NOT in pyproject.toml
10     # [project.dependencies]. Their CPU/CUDA wheels need an out-of-band index
11     # (--index-url / --extra-index-url) that PEP 621 cannot express. Install
12     # them separately matching your hardware:
13     #   NVIDIA GPU: pip install torch torchvision --index-url
https://download.pytorch.org/whl/cu124
14     #   CPU only:   pip install torch torchvision --index-url
https://download.pytorch.org/whl/cpu
15     # CI installs CPU torch on its own line in .github/workflows/ci.yml.
16     torch>=2.12.0
17     torchvision>=0.27.0
18     supervision>=0.21.0
19     # D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the standalone
20     # `trackers` package. The PyPI wheel is BROKEN (ships metadata only, no source ? see
21     # https://github.com/roboflow/trackers/issues/474), so install from git until fixed:
22     #   pip install "trackers @ git+https://github.com/roboflow/trackers.git@2.5.0"
23     # (Apache-2.0; depends on supervision >=0.26.1, so it composes cleanly.)
24     # NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
25     # PIN-DUPLICATION (mirrors the torch carve-out): this exact ref is ALSO hard-coded in
26     # .github/workflows/ci.yml + nightly.yml (full-suite lanes) and
deploy/digitalocean/bootstrap.sh
27     # + deploy/cloudrun/Dockerfile ? edit ALL of them in lockstep on a bump, or the
lanes/boxes
28     # silently resolve different versions.
29     trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
30     scipy>=1.10
31
32     # Per-video observability reports (SOFT dependency: the pipeline degrades to a
33     # no-op if absent). Shared, permissively-licensed report toolkit. Pin a tag once
34     # the private repo is published, e.g.:
35     #   obsreport @ git+ssh://git@github.com/avidullu/sports-obsreport.git@v0.1.3
36     # For local development install it editable from a sibling checkout:
37     #   pip install -e ../sports-obsreport --no-build-isolation
38

```

16. user (2026-07-02T02:48:47.007Z)

```

.gitignore: Unicode text, UTF-8 text, with CRLF line terminators
8:.env
9:*.env
10:.env.*
11:# ...but deploy/gcp/buckets.env is the TRACKED single-source-of-truth for PUBLIC project +
bucket
12:# MUST be committed. Keep real secret .env files ignored above.
13:# MUST be committed. Keep real secret .env files ignored above.
14:!deploy/gcp/buckets.env
52:# Local generated artifacts (review reels, eval CSVs, e2e/proof outputs) -- regenerable,
never committed
53:artifacts/
69:# Throwaway repro scripts + PR-body / handoff scratch notes (already excluded from
71:scratch/
---tracked artifacts/scratch---
scratch/analyze_motion.py
scratch/clean_state.py
scratch/diagnose_rally.py
scratch/gemini_video_debugger.py
scratch/list_models.py
scratch/test_ai_segmenter.py
scratch/test_thresholds.py

```

```
---tracked env/config---
config.json
---untracked status---
!! artifacts/
!! scratch/ABLATION_FRAMEWORK_PLAN.md
!! scratch/GATE_AB_NUMBERS_AND_IMPLICATIONS.md
!! scratch/HIGH_RISK_REMEDIATION_PLAN.md
!! scratch/OWNED_MODEL_PHASE0_PLAN.md
!! scratch/PACKAGING_PLAN.serving-wip.md
!! scratch/PERSONALIZATION_FEATURE_DESIGN.md
!! scratch/PERSONALIZATION_PIVOT_ANALYSIS.md
!! scratch/PR1_BODY.md
!! scratch/PR1_HANDOFF.md
```

17. user (2026-07-02T02:48:47.412Z)

```
deploy\digitalocean\gpu_setup.sh
deploy\digitalocean\mount_scratch.sh
deploy\digitalocean\bootstrap.sh
deploy\digitalocean\README.md
```

18. user (2026-07-02T02:49:08.782Z)

```
{
  "sport": "badminton",
  "ai_provider": "gemini",
  "ai_model": "gemini-flash-latest",
  "validation": {
    "min_resolution_width": 1280,
    "min_resolution_height": 720,
    "min_fps": 29.9,
    "min_blur_threshold": 20.0,
    "max_scene_cuts_per_minute": 0.5,
    "relevance": {
      "court_green_lower": [
        35,
        40,
        40
      ],
      "court_green_upper": [
        85,
        255,
        255
      ],
      "court_pixels_min_ratio": 0.05
    }
  },
  "indexing": {
    "default_segmenter": "gemini",
    "use_gpu": true,
    "motion_threshold": 0.08,
    "min_segment_duration": 3.0,
    "dead_time_merge_gap": 2.0,
    "chunk_duration": 90,
    "chunk_overlap": 10,
    "detector_model": "fasterrcnn_resnet50_fpn",
    "detector_score_threshold": 0.5,
    "yolo_velocity_threshold": 0.15,
    "yolo_frame_skip": 3,
    "yolo_tracker": "bytetrack.yaml",
    "yolo_prefetch_workers": 2,
    "min_action_window_duration": 2.0,
    "log_yolo_candidates": true,
    "store_yolo_candidates": true,
    "ai_handoff_padding": 2.0,
```

```

"ai_max_workers": 1,
"tracknet": {
  "wsl_distro": "Ubuntu",
  "repo_dir": "~/models/TrackNetV4",
  "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
  "conda_env": "TrackNetV4",
  "weights_path": "",
  "queue_length": 5,
  "stage_in_wsl": true,
  "wsl_stage_dir": "~/clips",
  "timeout_sec": 1800,
  "velocity_threshold": 0.02
},
"wasb": {
  "wsl_distro": "Ubuntu",
  "repo_dir": "~/models/WASB-SBDT",
  "conda_sh": "~/miniconda3/etc/profile.d/conda.sh",
  "conda_env": "wasb",
  "weights_path": "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar",
  "sport": "badminton",
  "wsl_stage_dir": "~/clips",
  "timeout_sec": 1800,
  "velocity_threshold": 0.02
}
},
"output": {
  "default_highlights_name": "highlights.mp4",
  "db_name": "sports_indexer.db"
},
"stitching": {
  "begin_padding": 0.5,
  "end_padding": 0.5,
  "use_cache": false,
  "min_shot_count": 1,
  "watermark": {
    "enabled": true,
    "text": "Generated by Khelsutra.guru",
    "position": "bottom-right",
    "opacity": 0.85,
    "logo_path": "",
    "font_path": ""
  }
}
}
}

```

19. user (2026-07-02T02:49:10.641Z)

```

.github\workflows\nightly.yml:131:      - name: Install trackers (git-only dep ? PyPI wheel is
broken)
.github\workflows\nightly.yml:132:      # D13/P12: ByteTrack moved to the standalone
`trackers` package (git-only;
.github\workflows\nightly.yml:135:      run: python -m pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
.github\workflows\ci.yml:124:      - name: Install trackers (git-only dep ? PyPI wheel is
broken)
.github\workflows\ci.yml:127:      # source ? roboflow/trackers#474), so it MUST come from
git. PEP 621 can't
.github\workflows\ci.yml:131:      # NOTE: roboflow/trackers uses bare version tags (2.5.0),
NOT v-prefixed.
.github\workflows\ci.yml:132:      run: python -m pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
.github\workflows\ci.yml:180:      # D13/P12 carve-out: `trackers` (Apache-2.0; ByteTrack's new
home) is also git-only and thus
requirements.txt:21:# https://github.com/roboflow/trackers/issues/474), so install from git
until fixed:

```

```

requirements.txt:22:# pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
requirements.txt:24:# NOTE: roboflow/trackers uses bare version tags (2.5.0), NOT v-prefixed.
requirements.txt:29:trackers @ git+https://github.com/roboflow/trackers.git@2.5.0
deploy\digitalocean\bootstrap.sh:42:# `trackers` package. Git-only dep (PyPI wheel is metadata-
only ? roboflow/trackers#474),
deploy\digitalocean\bootstrap.sh:46: pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0"
deploy\cloudrun\Dockerfile:58:# D13/P12: `trackers` (ByteTrack's new home) is a git-only dep
(broken PyPI wheel ?
deploy\cloudrun\Dockerfile:59:# roboflow/trackers#474) so it's NOT pulled by `pip install .`;
install it out-of-band before
deploy\cloudrun\Dockerfile:62:    && python3 -m pip install "trackers @
git+https://github.com/roboflow/trackers.git@2.5.0" \
backend\ui\assets\index-ukuS9wxa.js:60:[Omitted long matching line]
docs\CODE_CLEANUP_AUDIT_2026-06-24.md:200:| **P12** | **D13: ByteTrack deprecation** ? migrated
`sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel broken) | P7 | No | ? |
[#386](https://github.com/Khelsutra/badminton-highlight-indexer/pull/386) |
docs\CODE_CLEANUP_AUDIT_2026-06-24.md:246:| ~4~ | ~**D13 / P12**~ | ~ByteTrack deprecation
migration~ ? **? DONE.** `sv.ByteTrack` ? `trackers.ByteTrackTracker` (git dep; PyPI wheel
broken ? filed Roboflow issue). |
docs\CODE_CLEANUP_AUDIT_2026-06-24.md:282:[Omitted long matching line]
tests\test_person_detector.py:210:    Skipped when the `trackers` package isn't installed (it's
a git-only dep whose

```

20. user (2026-07-02T02:49:27.183Z)

Testing the rally pipeline on DigitalOcean

```

> **? DEPRECATED (ADR-013, 2026-06-20) ? DigitalOcean is DROPPED for compute.** Use
> **[`deploy/gcp/README.md`](../gcp/README.md)** ? GCP is the sole compute stack (DO GPU isn't
enabled
> on the account + isn't cleanly credit-funded; Paperspace is subscription-gated). Kept for
history.
> **NOTE:** the scripts here ? `bootstrap.sh`, `mount_scratch.sh`, `gpu_setup.sh` ? are
> **provider-agnostic** Linux-GPU setup and are **still reused by the GCP runbook** ; only the
> DO-specific *provisioning* (doctl droplet create, Spaces) is retired. (Optional later cleanup:
move
> them to `deploy/common/`.)

```

A storage-free, secret-free way to get the pipeline running on a DigitalOcean (or Paperspace) machine ? the **compute** half of [`docs/archives/past_projects/DECOUPLED_COMPUTE/`](`../docs/archives/past_projects/DECOUPLED_COMPUTE/README.md`).
The object-store **storage shim** is intentionally out of scope here (owner is finalizing it); this runbook only proves **compute** works on the box.

```

===
===

```

```

OPEN | ci: single-source the shared offline guard lanes + coverage floor via a workflow_call
reusable
OPEN | [cost] /api/process re-runs the paid segmenter for an already-processed video ?
refresh/"Try again" re-bills Gemini
OPEN | [safety] Content-safety pre-flight litmus: flag non-sports / NSFW video BEFORE sending it
to Gemini

```

21. user (2026-07-02T02:49:27.192Z)

```

1      ""Cloudflare Access edge-auth ? verify the Cf-Access JWT, derive the tenant ``user_id``
(M9, D9).
2
3      **DEFAULT-OFF:** when ``security.edge_auth_enabled`` is False the resolver is a strict
NO-OP (returns
4      ``None`` = the local single-user) ? nothing about the existing local path changes. When

```

```

enabled, the
5     ``Cf-Access-Jwt-Assertion`` header is **required** and **verified** (RS256) against the
team JWKS ?
6     signature + exp + issuer + audience ? so a direct hit to the Cloud Run URL **can't
spooof** the
7     identity header (the explicit D9 anti-spoof requirement). Returns the verified
``user_id`` (the
8     token's ``email``, else ``sub``).
9
10    ``jwt`` (``PyJWT[crypto]``) is **lazy-imported** so the base runtime works without it
(install the
11    ``auth`` extra for hosted/multi-tenant). The JWKS fetch is **injectable**, so the whole
verifier is
12    unit-tested **offline** with an in-test RSA keypair + a fake JWKS ? no network, no
Cloudflare.
13    """
14
15    from __future__ import annotations
16
17    import json
18    import time
19    import urllib.request
20    from typing import Any, Callable, Mapping, Optional
21
22    # Cloudflare Access sends the signed assertion in this header (it terminates at the CF
edge).
23    CF_ACCESS_HEADER = "Cf-Access-Jwt-Assertion"
24    _JWKS_PATH = "/cdn-cgi/access/certs"
25    _JWKS_TTL_SEC = (
26        600.0 # re-fetch the team JWKS at most every 10 min (keys rotate slowly)
27    )
28
29    JwksFetcher = Callable[[str], dict]
30
31
32    class EdgeAuthError(Exception):
33        """A required Cf-Access token was missing or failed verification ? the caller
returns 401/403."""
34
35
36    # module-level JWKS cache: team_domain -> (fetched_at, jwks_dict). Production only;
tests inject.
37    _jwks_cache: dict[str, tuple[float, dict]] = {}
38
39
40    def _issuer(team_domain: str) -> str:
41        return team_domain.rstrip("/")
42
43
44    def _default_jwks_fetcher(team_domain: str) -> dict:
45        """Fetch the team's public JWKS (``<team_domain>/cdn-cgi/access/certs``). Network ?
only the
46        production path reaches here; tests pass an injected fetcher."""
47        url = _issuer(team_domain) + _JWKS_PATH
48        with urllib.request.urlopen(url, timeout=5) as resp: # noqa: S310 (https team
domain)
49            data: Any = json.load(resp)
50            if not isinstance(data, dict) or "keys" not in data:
51                raise EdgeAuthError(f"unexpected JWKS shape from {url}")
52            return data
53
54
55    def _get_jwks(team_domain: str, fetcher: Optional[JwksFetcher]) -> dict:
56        """Return the JWKS for ``team_domain``, cached with a TTL. An injected ``fetcher``
(tests)

```

```

57     bypasses the cache so each test is deterministic."""
58     if fetcher is not None:
59         return fetcher(team_domain)
60     now = time.time()
61     cached = _jwks_cache.get(team_domain)
62     if cached and (now - cached[0]) < _JWKS_TTL_SEC:
63         return cached[1]
64     jwks = _default_jwks_fetcher(team_domain)
65     _jwks_cache[team_domain] = (now, jwks)
66     return jwks
67
68
69     def verify_cf_access_token(
70         token: str, *, team_domain: str, aud: str, jwks: dict
71     ) -> str:
72         """Verify a Cf-Access JWT against ``jwks`` (RS256; checks sig + exp + issuer +
73         audience) and
74         return the user_id (``email`` else ``sub``). Raises ``EdgeAuthError`` on ANY
75         failure."""
76         try:
77             import jwt
78             from jwt import PyJWKSet
79         except (
80             Exception
81         ) as exc: # pragma: no cover - only when the auth extra isn't installed
82             raise EdgeAuthError(
83                 "PyJWT[crypto] is required for edge auth ? install the 'auth' extra "
84                 "(pip install -e .[auth]) on a host with edge_auth_enabled."
85             ) from exc
86         try:
87             kid = jwt.get_unverified_header(token).get("kid")
88             if not kid:
89                 raise EdgeAuthError("token header has no kid")
90             signing_key = PyJWKSet.from_dict(jwks)[kid]
91             claims = jwt.decode(
92                 token,
93                 signing_key.key,
94                 algorithms=["RS256"],
95                 audience=aud,
96                 issuer=_issuer(team_domain),
97                 options={"require": ["exp", "iss", "aud"]},
98             )
99         except EdgeAuthError:
100             raise
101         except KeyError as exc: # kid not in the JWKS
102             raise EdgeAuthError(f"no JWKS key for kid {exc}") from exc
103         except (
104             Exception
105         ) as exc: # jwt.* errors: bad sig / expired / wrong aud|iss / malformed
106             raise EdgeAuthError(f"Cf-Access token verification failed: {exc}") from exc
107
108         user = claims.get("email") or claims.get("sub")
109         if not user:
110             raise EdgeAuthError("verified token carries no email/sub claim")
111         return str(user)
112
113     def resolve_tenant(
114         headers: Mapping[str, str],
115         config: Any,
116         *,
117         jwks_fetcher: Optional[JwksFetcher] = None,
118     ) -> Optional[str]:
119         """Resolve the tenant ``user_id`` for a request, or ``None`` when edge auth is OFF.

```

```

120
121     DEFAULT-OFF (`security.edge_auth_enabled=False`) ? ``None`` (the local single-
user; unchanged).
122     Enabled ? require + verify the Cf-Access header; raise ``EdgeAuthError`` (? 401/403)
on any miss.
123     ``jwks_fetcher`` is injectable for offline tests.
124     """
125     sec = getattr(config, "security", None)
126     if not sec or not getattr(sec, "edge_auth_enabled", False):
127         return None # DEFAULT-OFF ? no tenant identity, byte-identical to today
128
129     # Strip so a whitespace-only value counts as unset ? matching the boot-time exposure
check in
130     # config/startup.py (EXPOSED_EDGE_AUTH_INCOMPLETE); one semantics for "configured",
both places.
131     team_domain = (getattr(sec, "cf_team_domain", "") or "").strip()
132     aud = (getattr(sec, "cf_access_aud", "") or "").strip()
133     if not team_domain or not aud:
134         raise EdgeAuthError(
135             "edge_auth_enabled but security.cf_team_domain / cf_access_aud are not
configured"
136         )
137
138     # Headers are case-insensitive over HTTP; Starlette's Headers handles that, but
support a plain
139     # dict too (tests) by trying both casings.
140     token = headers.get(CF_ACCESS_HEADER) or headers.get(CF_ACCESS_HEADER.lower())
141     if not token:
142         raise EdgeAuthError(
143             f"missing {CF_ACCESS_HEADER} header (request did not pass Cloudflare
Access)"
144         )
145
146     jwks = _get_jwks(team_domain, jwks_fetcher)
147     return verify_cf_access_token(token, team_domain=team_domain, aud=aud, jwks=jwks)
148

```

22. user (2026-07-02T02:49:38.217Z)

```

backend\main.py:731:     SELF-HOST posture only (D7): refused with 403 when
``security.allowed_video_root`` is set ? the
backend\main.py:747:     if getattr(sec, "allowed_video_root", None):
backend\main.py:752:         "(security.allowed_video_root is set); upload to local
instead."
backend\main.py:1373:         getattr(request.app.state.config, "security", None),
"allowed_video_root", None
backend\main.py:1388:     # input: its null-byte guard always applies, and a configured
allowed_video_root (hosted mode)
backend\main.py:1797:     auth on without the ``allowed_video_root`` jail / CF team+AUD). A
strict no-op when no
backend\config\models.py:330:     allowed_video_root: Optional[str] = None
backend\config\models.py:334:     # arrive with PR-3 identity scoping. ? In hosted mode, set
allowed_video_root to
backend\config\startup.py:22:rest of the posture at **boot**: the ``allowed_video_root`` path-
jail is set (else an authenticated
backend\config\startup.py:107:     # a tenant can point the engine at an ARBITRARY local file
(allowed_video_root unset = "any local
backend\config\startup.py:109:     root = getattr(sec, "allowed_video_root", None)
backend\config\startup.py:115:         "security.edge_auth_enabled is ON (engine exposed)
but security.allowed_video_root is "
backend\config\startup.py:117:         "Set allowed_video_root to a jail directory before
exposing the engine.",
backend\utils\paths.py:58:def validate_input_path(path: str, allowed_root: Optional[str] = None)
-> None:

```

23. user (2026-07-02T02:49:43.633Z)

```
1      """Path-safety helpers for the API boundary (untrusted request fields)."""
2
3      import os
4      from typing import Any, Optional
5
6
7      def output_base(config: Any) -> str:
8          """The configured base dir for a segmenter's LOCAL scratch/staging artifacts
9              (COMPUTE_DECOUPLED_SERVING M3). Returns ``output.output_dir`` (default ``"output"``)
10 so the
11         chunk-staging / AI-handoff / temp dirs follow ``--output-dir`` /
12 ``output.output_dir`` instead of
13         a hardcoded ``./output``. The DEFAULT is ``"output"`` ? byte-identical to the prior
14 literal.
15         This is the LOCAL output dir, deliberately NOT the storage ``workspace_root`` (which
16 may be a
17         ``gs://`` URI you can't ``mkdirs``). Accepts the typed AppConfig or the legacy
18 dict-shim
19         (``.get("output", {})).get("output_dir")``); any miss or empty value falls back to
20 ``"output"``. """
21     try:
22         out = (
23             config.output.output_dir
24             if not isinstance(config, dict)
25             else config.get("output", {}).get("output_dir")
26         )
27     except Exception:
28         out = None
29     return out or "output"
30
31
32 def safe_output_filename(name: str) -> str:
33     """Validate an output filename is a BARE name (no dir part, no traversal, not
34 absolute).
35     ``os.path.join(output_dir, name)`` silently discards ``output_dir`` when ``name`` is
36 absolute,
37 and ``..`` segments escape it ? so an unchecked ``output_filename`` is an arbitrary-
38 write
39 primitive. Returns the name unchanged when safe; raises ValueError otherwise.
40 """
41     if not name or not name.strip():
42         raise ValueError("output_filename must not be empty")
43     if "\x00" in name:
44         raise ValueError("output_filename must not contain a null byte")
45     if "/" in name or "\\" in name:
46         raise ValueError(f"output_filename must not contain a path separator: {name!r}")
47     if os.path.isabs(name) or name != os.path.basename(name) or name in (".", ".."):
48         raise ValueError(f"output_filename must be a bare filename, got: {name!r}")
49     return name
50
51
52 def resolve_under_root(path: str, root: str) -> str:
53     """Return the real path, asserting it is contained under ``root`` (raises ValueError
54 if not). """
55     rp = os.path.realpath(path)
56     rr = os.path.realpath(root)
57     try:
58         contained = os.path.commonpath([rp, rr]) == rr
59     except ValueError: # different drives on Windows
60         contained = False
61     if not contained:
```

```

54         raise ValueError(f"path {path!r} escapes the allowed root {root!r}")
55     return rp
56
57
58     def validate_input_path(path: str, allowed_root: Optional[str] = None) -> None:
59         """Reject null bytes and (when an allowed_root is configured) out-of-root inputs.
60
61         With no allowed_root (default, single-user local mode) only the null-byte check
62         applies,
63         so the local 'point it at any of my videos' workflow is preserved. Setting an
64         allowed_root (a future hosted/tenant deployment) enforces containment.
65         """
66         if "\x00" in path:
67             raise ValueError("video_path must not contain a null byte")
68         if allowed_root:
69             resolve_under_root(path, allowed_root)

```

24. user (2026-07-02T02:49:44.530Z)

```

1360     client gets an immediate 4xx; everything slow ? including MD5 hashing of the file ?
1361     runs on the worker (DEFERRED_HARDENING_PLAN #16). The row is created 'queued'; the
1362     worker drains it via `claim_due_job` (EXECUTION_POLICY.md P3 self-scheduling loop).
1363     """
1364     # M9 (D9): edge-auth tenant resolution. DEFAULT-OFF ? owner_id None (the local
1365     single-user,
1366     # unchanged). When edge_auth_enabled, the Cf-Access JWT is required + verified (a
1367     missing or
1368     # spoofed identity header fails here with 401, before any work) and tags the job's
1369     owner_id.
1370     try:
1371         owner_id = edge_auth.resolve_tenant(request.headers, request.app.state.config)
1372     except edge_auth.EdgeAuthError as e:
1373         raise HTTPException(status_code=401, detail=f"edge auth: {e}") from e
1374
1375     allowed_root = getattr(
1376         getattr(request.app.state.config, "security", None), "allowed_video_root", None
1377     )
1378     try:
1379         validate_input_path(req.video_path, allowed_root=allowed_root)
1380         # M2: resolve the input once (path or storage URI) ? raises ValueError for an
1381         unsupported
1382         # scheme, which we map to a clean 400 (an unknown scheme must never 500).
1383         input_ref = storage.resolve_input_ref(
1384             req.video_path, getattr(request.app.state.config, "storage", None)
1385         )
1386     except ValueError as e:
1387         raise HTTPException(status_code=400, detail=str(e)) from e
1388     # The local-existence fast-fail applies only to a LOCAL input, and stats the
1389     RESOLVED local path
1390     # (input_ref.key ? e.g. local:/// stripped to the bare path), not the raw URI
1391     string. A
1392     # bucket/Drive URI can't be cheaply os.path.exists()'d ? its existence is verified
1393     when the worker
1394     # fetches it (a fetch miss fails the job loudly). validate_input_path above still
1395     runs for every
1396     # input: its null-byte guard always applies, and a configured allowed_video_root
1397     (hosted mode)
1398     # rejects a non-local URI as out-of-root.
1399     if input_ref.backend == "local" and not os.path.exists(input_ref.key):
1400         raise HTTPException(
1401             status_code=404, detail=f"Video file not found at '{req.video_path}'"
1402         )
1403     # P2 (D11): a bucket/Drive source is fetched to a scratch copy on the worker; fail
1404     fast with 507

```

```

1395         # if this box can't hold that copy. Best-effort ? an unstatable remote
(network/SDK) or an
1396         # unreadable temp fs skips the guard (measuring must not deny a legit job); the
fetch itself
1397         # still fails loudly downstream if space genuinely runs out. Local inputs need no
copy ? skipped.
1398         if input_ref.backend != "local":
1399             need = _remote_input_size_bytes(
1400                 input_ref, getattr(request.app.state.config, "storage", None)
1401             )
1402             require_free_bytes(tempfile.gettempdir(), need, stage="fetch")
1403         if req.segmenter and not segmenter_registry.get(req.segmenter):
1404             available = list(segmenter_registry.list_available().keys())
1405             raise HTTPException(
1406                 status_code=400,
1407                 detail=f"Segmenter '{req.segmenter}' not found. Available segmenters:
{available}",
1408             )
1409
1410         job_id = uuid.uuid4().hex
1411         if not request.app.state.db.create_job(
1412             job_id,
1413             req.video_path,
1414             req.segmenter,
1415             req.player_pool,
1416             req.max_frames,
1417             owner_id=owner_id,
1418             locale=req.locale,
1419             compute=req.compute,

```

25. user (2026-07-02T02:49:55.154Z)

```

deploy\gcp\README.md:93:> **always bakes in the Ops Agent** (`--metadata-from-file startup-
script=deploy/gcp/ops-agent-startup.sh`)
deploy\gcp\README.md:144:> `--metadata=startup-script` so observability is automatic on every
box.
deploy\gcp\README.md:151:> driver: either add `--metadata=install-nvidia-driver=True` style
startup, or after SSH run GCP's
deploy\gcp\README.md:187:mkdir -p /root/.keys && printf '%s' '<ROTATED_GEMINI_KEY>' >
/root/.keys/GEMINI_API_KEY && chmod 600 /root/.keys/GEMINI_API_KEY
deploy\gcp\ops-agent-startup.sh:5:# safe. Baked into EVERY rally GPU VM by
deploy/gcp/create_gpu_vm.sh (`--metadata-from-file
deploy\gcp\create_gpu_vm.sh:3:# spin up a rally GPU box ? it bakes the ops-agent startup-script
into the VM metadata, attaches the
deploy\gcp\create_gpu_vm.sh:49:         --metadata-from-file=startup-script="$STARTUP" 2>"$ERR";
then
deploy\digitalocean\README.md:29:| 6 | **`GEMINI_API_KEY`** | Google AI Studio (or Vertex) | the
Gemini segmenter | **later** ? only to test the *Gemini* path; the stub/WASB path needs none.
**Rotate the old key that was shared in chat.** |
deploy\digitalocean\bootstrap.sh:42:# `trackers` package. Git-only dep (PyPI wheel is metadata-
only ? roboflow/trackers#474),
deploy\gcp\nightly_regression\launch.sh:60: echo "[dry-run] staged $TGZ_URI OK; would create
$GPU VM '$NAME' with run_on_vm.sh + metadata. Skipping VM create."
deploy\gcp\nightly_regression\launch.sh:64:# 2) Zone-sweep create the VM (mirrors
create_gpu_vm.sh) with run_on_vm.sh + the config metadata.
deploy\gcp\nightly_regression\launch.sh:74:         --metadata-from-file=startup-script="$STARTUP"
\
deploy\gcp\nightly_regression\launch.sh:75:         --metadata=repo-tgz-uri="$TGZ_URI",git-
sha="$SHA",report-bucket="$NIGHTLY_BUCKET",report-prefix="$REPORT_PREFIX",weights-
uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-
to="$EMAIL_TO" \
deploy\gcp\nightly_regression\README.md:140:| `reels=0`, log mentions GEMINI_API_KEY |
`skip_ai_handoff` not true ? ensure the manifest `config_overrides` has
`indexing.skip_ai_handoff: true` |
deploy\gcp\nightly_regression\run_on_vm.sh:4:# an interactive session). Reads its config from

```

```
instance metadata (set by launch.sh).
deploy\gcp\nightly_regression\run_on_vm.sh:21:
"http://metadata.google.internal/computeMetadata/v1/$1"; }
deploy\gcp\nightly_regression\run_on_vm.sh:70:if [ -z "$REPO_TGZ" ]; then echo "FATAL: repo-tgz-
uri metadata missing ? nothing to run."; exit 1; fi
```

26. user (2026-07-02T02:49:56.741Z)

```
scripts\run_phase3_eval.py:14:Full corpus with the two-stage detector (needs GEMINI_API_KEY;
costs API calls):
scripts\run_phase3_eval.py:31:load_dotenv() # make GEMINI_API_KEY (etc.) available to the AI-
backed segmenters
scripts\run_phase3_eval.py:65:          "yolo_hybrid/gemini need GEMINI_API_KEY and cost API
calls.",
tools\generate_highlights.py:109:      # Load .env (e.g. GEMINI_API_KEY) so the Gemini handoff
works when the runner is invoked
```

27. user (2026-07-02T02:50:06.426Z)

```
1      #!/usr/bin/env bash
2      # Launch ONE nightly engine-regression run on an ephemeral GCP GPU VM, then return. The
VM runs
3      # run_on_vm.sh as its startup-script (regression ? email ? SELF-DELETE), so this just
stages the repo
4      # and kicks the VM off. Driver-agnostic: call it from Cloud Scheduler?Cloud Run job, a
cron on any
5      # always-on box, the owner's machine, or a Claude routine. Needs gcloud auth + the repo
checkout.
6      #
7      #   bash deploy/gcp/nightly_regression/launch.sh                # L4, stages HEAD,
kicks the VM
8      #   RALLY_GPU=t4 RALLY_REF=$(git rev-parse HEAD) bash
deploy/gcp/nightly_regression/launch.sh
9      #   RALLY_DRY_RUN=1 bash deploy/gcp/nightly_regression/launch.sh # stage only, DON'T
create the VM (smoke test)
10     #
11     # Works BOTH from a local git checkout (archives REF) AND from the Cloud Run launcher
container, which has
12     # no .git (it tars the baked tree + takes the SHA from $RALLY_GIT_SHA ? set by
register_scheduler.sh).
13     #
14     # ? The VM-side flow is NOT covered by CI. Tail: gcloud compute instances get-serial-
port-output <name> --zone=<z>.
15     set -uo pipefail
16
17     # Centralized project + bucket names (single source of truth ? deploy/gcp/buckets.env).
Now honors
18     # RALLY_CORPUS_BUCKET (canonical) as well as the legacy RALLY_BUCKET / RALLY_GCS_BUCKET.
19     source "$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/../buckets.env"
20
21     PROJECT="$RALLY_GCP_PROJECT"
22     CORPUS_BUCKET="$RALLY_CORPUS_BUCKET" # PERMANENT inputs: clips/labels/weights (no
TTL)
23     NIGHTLY_BUCKET="$RALLY_NIGHTLY_BUCKET" # EPHEMERAL outputs: reports + repo tarball (30d
TTL)
24     SA="rally-worker@${PROJECT}.iam.gserviceaccount.com"
25     GPU="${RALLY_GPU:-l4}"
26     NAME="${RALLY_VM_NAME:-rally-nightly-$(date -u +%Y%m%d-%H%M%S)}" # to-the-second ?
same-day re-runs don't collide
27     REF="${RALLY_REF:-HEAD}"
28     WEIGHTS_URI="$RALLY_WEIGHTS_URI" # gs://<corpus>/models/wasb_badminton_best.pth.tar
(derived in buckets.env; validated 2026-06-22)
29     REPORT_PREFIX="${RALLY_REPORT_PREFIX:-reports}"
30     DO_EMAIL="${RALLY_EMAIL:-true}"
31     SMTP_SECRET="${NIGHTLY_SMTP_SECRET:-projects/$PROJECT/secrets/nightly-smtp-
```

```

pass/versions/latest}"
32     SMTP_USER="${NIGHTLY_SMTP_USER:-}"
33     EMAIL_TO="${NIGHTLY_EMAIL_TO:-avi.dullu@gmail.com}"
34     DRY_RUN="${RALLY_DRY_RUN:-}"
35     HERE="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)"
36     ROOT="$(cd "$HERE/../../.." && pwd)" # repo root (deploy/gcp/nightly_regression ? 3
up)
37     STARTUP="$HERE/run_on_vm.sh"
38     export CLOUDSDK_CORE_DISABLE_PROMPTS=1
39
40     ZONES="${RALLY_GCP_ZONES:-asia-south1-a asia-south1-b asia-south1-c asia-southeast1-a
asia-southeast1-b asia-southeast1-c}"
41     if [ "$GPU" = "l4" ]; then MACHINE="g2-standard-4"; ACCEL=(); else
MACHINE="n1-standard-8"; ACCEL=(--accelerator=type=nvidia-tesla-t4,count=1); fi
42
43     # 1) Stage the repo to the EPHEMERAL nightly bucket (30d TTL; no GitHub auth on the VM).
44     #     Create the bucket first if missing: bash
deploy/gcp/nightly_regression/create_nightly_bucket.sh
45     #     Two modes: a git checkout archives the tracked tree @ REF; the Cloud Run container
(no .git) tars
46     #     the baked tree and takes the SHA from $RALLY_GIT_SHA / a baked GIT_SHA file.
47     SHA="${RALLY_GIT_SHA:-$( (cd "$ROOT" && git rev-parse "$REF" 2>/dev/null) || cat
"$ROOT/GIT_SHA" 2>/dev/null || echo unknown )}"
48     TGZ_URI="gs://$NIGHTLY_BUCKET/repo/${SHA}.tgz"
49     echo "== staging repo $SHA -> $TGZ_URI =="
50     TMP_TGZ="$(mktemp --suffix=.tgz)"
51     if (cd "$ROOT" && git rev-parse --git-dir >/dev/null 2>&1); then
52         ( cd "$ROOT" && git -c core.autocrlf=false archive --format=tar.gz -o "$TMP_TGZ"
"$REF" ) # local checkout
53     else
54         tar czf "$TMP_TGZ" -C "$ROOT" --exclude='./.git' .
# container: baked tree
55     fi
56     gcloud storage cp "$TMP_TGZ" "$TGZ_URI" --project="$PROJECT"
57     rm -f "$TMP_TGZ"
58
59     if [ -n "$DRY_RUN" ]; then
60         echo "[dry-run] staged $TGZ_URI OK; would create $GPU VM '$NAME' with run_on_vm.sh +
metadata. Skipping VM create."
61         exit 0
62     fi
63
64     # 2) Zone-sweep create the VM (mirrors create_gpu_vm.sh) with run_on_vm.sh + the config
metadata.
65     ERR="$(mktemp)"
66     for Z in $ZONES; do
67         echo "== creating $NAME ($GPU) in $Z =="
68         if gcloud compute instances create "$NAME" --project="$PROJECT" --zone="$Z" \
69             --machine-type="$MACHINE" "${ACCEL[@]}" \
70             --maintenance-policy=TERMINATE --provisioning-model=STANDARD \
71             --image-family=common-cu129-ubuntu-2204-nvidia-580 --image-project=deeplearning-
platform-release \
72             --boot-disk-size=120GB --boot-disk-type=pd-balanced \
73             --service-account="$SA" --scopes=cloud-platform \
74             --metadata-from-file=startup-script="$STARTUP" \
75             --metadata=repo-tgz-uri="$TGZ_URI",git-sha="$SHA",report-
bucket="$NIGHTLY_BUCKET",report-prefix="$REPORT_PREFIX",weights-
uri="$WEIGHTS_URI",email="$DO_EMAIL",smtp-secret="$SMTP_SECRET",smtp-user="$SMTP_USER",email-
to="$EMAIL_TO" \
76             2>"$ERR"; then
77         echo "? launched $NAME in $Z ? it will run the regression, email $EMAIL_TO, and
self-delete."
78         echo "    serial log: gcloud compute instances get-serial-port-output $NAME --zone=$Z
--project=$PROJECT"
79         echo "    if it doesn't self-delete: gcloud compute instances delete $NAME --zone=$Z

```

```

-q"
80         rm -f "$ERR"; exit 0
81     fi
82     if grep -qiE "STOCKOUT|does not have enough" "$ERR"; then echo " (no $GPU capacity in
$Z ? next zone)"; continue; fi
83     echo " create failed:"; tail -4 "$ERR"; rm -f "$ERR"; exit 1
84 done
85     echo "? no $GPU capacity in any swept zone ($ZONES). Retry later or RALLY_GPU=t4."
86     rm -f "$ERR"; exit 1
87

```

28. user (2026-07-02T02:50:48.674Z)

```

41:from fastapi.staticfiles import StaticFiles
240:app.mount("/static", StaticFiles(directory=_STATIC_DIR), name="static")
255:    app.mount("/assets", StaticFiles(directory=_UI_ASSETS_DIR), name="assets")

```

29. user (2026-07-02T02:50:50.170Z)

No matches found

30. user (2026-07-02T02:50:51.853Z)

```

8615534 refactor: adopt approved layered directory structure
72f972e 2026-05-18 Checkpoint: Add debug trim feature, runner script, and motion analysis
scratch script

```

Coverage data

```

.coverage
.coverage.*
htmlcov/
.pytest_cache/

```

Local Claude Code config (machine-specific)

```

.claude/

```

Throwaway repro scripts + PR-body / handoff scratch notes (already excluded from ruff + pytest; never meant to be committed -- keep them out of `git status` too).

```

scratch/
.venv/

```

Private leak-scan name-list (real player/venue names for LOCAL `tools.leak_scan --na scanning -- NEVER committed; CI runs the name-free structural guard).

31. user (2026-07-02T02:51:18.624Z)

```

CLAUDE.md
README.md
__pycache__
artifacts
backend
backend.db
config.json
deploy
docs
eval_baselines
indexer.db
mypy.ini
output
pyproject.toml
pytest.ini

```

```

rally_index.db
requirements.txt
ruff.toml
run_all_tests.py
run_eval.py
run_regression.py
scratch
scripts
tests
tools
training
requirements.txt
deploy/gcp/create_gpu_vm.sh:20:# Centralized project + bucket names (single source of truth ?
deploy/gcp/buckets.env).
deploy/gcp/create_gpu_vm.sh:21:source "$(cd "$(dirname "${BASH_SOURCE[0]}")" &&
pwd)/buckets.env"
deploy/gcp/nightly_regression/create_nightly_bucket.sh:16:# Centralized project + bucket names
(single source of truth ? deploy/gcp/buckets.env).
deploy/gcp/nightly_regression/create_nightly_bucket.sh:17:source "$(cd "$(dirname
"${BASH_SOURCE[0]}")" && pwd)/../buckets.env"
deploy/gcp/nightly_regression/launch.sh:17:# Centralized project + bucket names (single source
of truth ? deploy/gcp/buckets.env). Now honors
deploy/gcp/nightly_regression/launch.sh:19:source "$(cd "$(dirname "${BASH_SOURCE[0]}")" &&
pwd)/../buckets.env"
deploy/gcp/nightly_regression/README.md:54:| **Buckets** | corpus (inputs) + `khehsutra-rally-
nightly` (outputs) | `RALLY_CORPUS_BUCKET` / `RALLY_NIGHTLY_BUCKET` ? the single source of truth
is `deploy/gcp/buckets.env` (sourced by every script). `RALLY_BUCKET`+ `RALLY_GCS_BUCKET`
remain accepted back-compat aliases for the corpus bucket |
deploy/gcp/nightly_regression/register_scheduler.sh:17:# Centralized project + bucket names
(single source of truth ? deploy/gcp/buckets.env).
deploy/gcp/nightly_regression/register_scheduler.sh:18:source "$(cd "$(dirname
"${BASH_SOURCE[0]}")" && pwd)/../buckets.env"
deploy/gcp/nightly_regression/setup_email.sh:16:# Centralized project + bucket names (single
source of truth ? deploy/gcp/buckets.env).
---
deploy/gcp/provision.sh:14:PROJECT="$RALLY_GCP_PROJECT"
deploy/gcp/create_gpu_vm.sh:23:PROJECT="$RALLY_GCP_PROJECT"
deploy/gcp/teardown.sh:25:PROJECT="$RALLY_GCP_PROJECT"
deploy/gcp/nightly_regression/register_scheduler.sh:20:PROJECT="$RALLY_GCP_PROJECT"
deploy/gcp/nightly_regression/create_nightly_bucket.sh:19:PROJECT="$RALLY_GCP_PROJECT"

```

32. user (2026-07-02T02:51:36.032Z)

```

---ignore-check---
.gitignore:44:*.db      backend.db
.gitignore:44:*.db      indexer.db
.gitignore:44:*.db      rally_index.db
.gitignore:47:output/   output
.gitignore:2:__pycache__/_pycache__
---gemini keyfile refs---
187:mkdir -p /root/.keys && printf '%s' '<ROTATED_GEMINI_KEY>' > /root/.keys/GEMINI_API_KEY &&
chmod 600 /root/.keys/GEMINI_API_KEY

```

33. user (2026-07-02T02:51:44.606Z)

```

deploy\gcp\nightly_regression\register_scheduler.sh:50:for ROLE in
roles/compute.instanceAdmin.v1 roles/iam.serviceAccountUser roles/run.invoker; do

```

34. user (2026-07-02T02:51:44.914Z)

```

1      # Cloud Build config for the Cloud Run GPU worker image (COMPUTE_DECOUPLED_SERVING M7).
2      #
3      # `gcloud builds submit --tag ? -f ?` does NOT work ? `gcloud builds submit` has no `-f`
flag, so a

```

```

4      # non-root Dockerfile needs a build CONFIG like this one. Build with:
5      #
6      #   gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project <PROJECT> .
7      #
8      # Override the region/repo/image with --substitutions=_REGION=?,_REPO=?,_IMAGE=? if
needed.
9      # PROJECT_ID is provided by Cloud Build automatically.
10     substitutions:
11         _REGION: asia-southeast1 # L4 GPU is region-limited (asia-southeast1 / europe-west1
/ europe-west4 / us-central1 / us-east4)
12         _REPO: rally
13         _IMAGE: rally-worker
14     steps:
15         - name: 'gcr.io/cloud-builders/docker'
16           args:
17             - build
18             - -t
19             - '${_REGION}-docker.pkg.dev/${PROJECT_ID}/${_REPO}/${_IMAGE}:latest'
20             - -f
21             - deploy/cloudrun/Dockerfile
22             - .
23     images:
24         - '${_REGION}-docker.pkg.dev/${PROJECT_ID}/${_REPO}/${_IMAGE}:latest'
25     options:
26         machineType: E2_HIGHCPU_8 # the CUDA + micromamba image is heavy; a bigger builder
keeps it ~8?10 min
27     timeout: 3600s
28

```

35. user (2026-07-02T02:53:37.550Z)

Structured output provided successfully